

Insider Trading, Earnings Quality, and Accrual Mispricing

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Abstract The paper provides evidence that the signal contained in insiders' trading behavior is useful in making refined assessments of earnings quality, and informative about the valuation implications of accruals. We find that income-increasing accruals and unexpected accruals have lower (higher) persistence when managers engage in abnormal selling (buying) suggesting that insider trading information is useful in assessing the quality of the non-cash components of earnings. We show that the accrual mispricing phenomenon observed in previous work is largely due to the mispricing of *positive* accruals. We find that investors price all positive accruals as if they were informative and that a subset of positive accruals is correctly priced. That is, (1) investors correctly price positive accruals that are likely to be informative because concurrently insiders engage in abnormal buying, and (2) investors price all positive accruals the same independently of insider trading. We find that the extent of the mispricing is greater when positive accruals occur concurrently with abnormal selling relative to cases where there is no trading. The extent of the mispricing and the magnitude of the one-year ahead returns (14.7 to 22 percent) to a trading strategy based on positive accruals and abnormal selling suggests that these accruals arise from opportunistic earnings management that is successful in misleading investors. By contrast, the smaller positive accrual mispricing when there is no insider trading is more likely related to either the complexity of the firms' accrual generating process (Thomas and Zhang (2001) or to earnings fixation (Sloan 1996). Our evidence thus suggests that opportunistic earnings management is a partial explanation for the accrual mispricing phenomenon.

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1. Introduction

This paper investigates whether the signal contained in insiders' trading behavior is useful in making refined assessments of earnings quality, and informative about the valuation implications of accruals.¹ Our work is motivated by the recent accumulation of evidence suggesting that investors (including sophisticated investors such as analysts and auditors) do not fully understand the persistence and valuation implications of accounting accruals.² Possible explanations for these findings include (1) the relation between the firm's accrual generating process and future earnings is sufficiently complex that investors fail to identify the transitory nature of the accruals [Thomas and Zhang (2001)], or (2) earnings have been managed opportunistically and investors fail to recognize until later periods that accruals are less persistent than operating cash flows [Bradshaw, Richardson and Sloan (2001), Hribar (2000), Tarpley (2000), Thomas and Zhang (2001), Xie (2001)]. Whether earnings have been managed or not, managers who make operational and accounting decisions that involve deferral, allocation and valuation choices are likely to possess private information regarding the nature and persistence of accounting accruals. Relying on extensive evidence that managers act as informed traders, selling (buying) in advance of stock price decreases (increases), we argue that insiders' trading behavior is informative *ex-ante* about their firm's earnings quality. Therefore, we examine: (1) whether firms in which the sign of the accruals and the direction of the insider trading are contradictory (confirmatory) have lower (higher) earnings quality, and (2) whether accrual mispricing is systematic or limited to firms with lower earnings quality.

The most prevalent explanation in previous work for accrual mispricing is that earnings management creates an information asymmetry between managers and investors because managers

¹ We define the *quality of current earnings* as the likelihood that the current earnings are sustainable in the future. Our tests of earnings quality are thus based on the relation between current earnings, its components, and future earnings.

² In his seminal study, Sloan (1996) shows that the cash flow component of earnings is more persistent than the accrual component of earnings and that investors do not understand the differential persistence and overweight the accrual component relative to the cash flow component. Several studies have followed Sloan and provided evidence consistent with accrual mispricing. For example, Bradshaw, Richardson and Sloan (2001), Hribar (2000) and Xie (2001) document that investors do not fully understand the valuation implications of accruals, unexpected accruals, and accrual components; Collins and Hribar (2000) show that accrual mispricing appears to be distinct from post-earnings announcement drift; Ali, Hwang and Trombley (2000) document that the accrual phenomenon is not related to firm size, analyst coverage, institutional ownership, volume or transaction costs.

exercise discretion opportunistically that investors fail to recognize. We believe there are two principal difficulties associated with this explanation. First, managers' accounting manipulations are largely unobservable. Consequently, as prior work acknowledges, inferences are critically dependent on the validity of constructs generated by various aggregate or specific accrual expectation models. Second, assuming that earnings have indeed been managed, prior work has not distinguished whether managers' exercise of discretion was intended to mislead or to inform. Thus, previous research reports that accruals are mispriced *on average*, but does not document whether market participants correctly price any accruals: Indeed, prior explanations require opportunistic earnings management to be pervasive.³

We believe that our research design choices provide empirical leverage for addressing these difficulties. First, we do not require that accruals be opportunistically managed for an information asymmetry between managers and investors to arise. We argue that asymmetric information with respect to the effect of strategic operating decisions on accruals can also lead to accrual mispricing. This type of information asymmetry is particularly plausible among young growth firms who have typically poorer information environments and where the accrual mispricing is most pronounced (Tarpley (2000)). Thus, we base part of our analysis on *raw* accruals and need not rely on accrual expectation models that may or may not isolate managerial discretion [Beneish (1997, 1998), McNichols (2000)].

Second, assuming that earnings are indeed managed, insider trading is an *observable* managerial action that can be informative about unobservable managers' accrual manipulations. By using insider trading, we exploit the fact that managers' discretionary accounting choices and their trading behavior are both signals of managers' private information. There is extensive evidence that

³ By opportunism, we mean managerial discretion over accruals that helps hide poor performance, and, by information, discretion that produces a more reliable measure of firm performance. Prior work has either not attempted or been unable to distinguish between opportunistic and informative earnings management. Holthausen (1990) observes that much prior work has predicated its conclusions on an opportunistic perspective for earnings management and has not tested the information perspective. In addition, recent attempts to distinguish between these two perspectives, by evaluating how estimates of managerial discretion is correlated with future stock market performance, have yielded mixed or conflicting evidence (Guay et al. (1996), Subramanyam (1996) and Teoh et al. (1998)).

managers' trades provide information to market participants and that they act as informed traders selling (buying) in advance of stock price decreases (increases) [e.g. Jaffe (1974); Penman (1982, 1985); Seyhun (1986)]. If, by managing earnings, managers reveal to investors their private information about the firm's prospects, we would expect the signal contained in their trading behavior to be consistent with the direction in which they exercise discretion over accruals. Thus, for example, income increasing earnings management, if informative about future growth, should not be accompanied by abnormally high levels of insider selling. Associating estimates of managers' discretion over accruals with the volume and direction of managers' trading activity and finding contradictory signals could, following Beneish (1999), inform us about the likelihood that earnings are opportunistically managed. This conjecture is supported by Beneish, Press, and Vargus (2001) who find that managers' accrual strategies and trading behavior are negatively correlated (positive accruals and abnormally high selling) in the period preceding technical default for those firms with the poorest post-default returns.

Our analyses are based on a sample of 21,678 observations, consisting of 3906 firms with December fiscal year-ends over the period 1985-1996.⁴ We conduct two types of analyses. First, our test of earnings quality are based on the relation between current and future earnings and more specifically on determining how the persistence of the accrual and unexpected accrual components of current earnings varies depending on the type of insider trading. Second, following Sloan (1996), we use Mishkin (1983) and hedge portfolio tests to examine whether insider trading is informative regarding the market pricing of accruals. Our application of the Mishkin test compares the market's valuation coefficient on accruals with the forecasting coefficient of accruals for one-year ahead earnings, with both equations conditioned on insider trading. The hedge portfolio test investigates the magnitude of the potential mispricing by evaluating abnormal returns to hedge portfolios formed

⁴ We constrain our sample to firms with December fiscal year-ends because our return tests are based on a trading strategy that requires an annual ranking of firms according to earnings management and insider trading activity. Our annual ranking occurs on April 30th, when annual financial statement data and insider trading data for the previous fiscal year are likely available for all sample firms. More important, at the time of the ranking the membership of the sample is known, thus making the ranking meaningful and the strategy implementable (see Holthausen (1983)).

on the basis of firms' accrual and insider trading properties.

Our findings are as follows. First, to benchmark our results, we reproduce Sloan's tests with our sample and find similar results. Second, since our analysis of insider trading requires directional predictions, we also reproduce Sloan's tests after partitioning accruals into positive and negative accruals: we find that positive accruals are mispriced but negative accruals are not. Third, we find that the persistence of income increasing accruals and unexpected accruals is significantly lower for firms with abnormal insider selling while firms with abnormal insider buying exhibit significantly greater earnings persistence. For example, the persistence of positive current (unexpected current) accruals is 0.754 (0.761) in firms with no insider trading activity, increases to 0.854 (0.976) in firms where accruals are accompanied by abnormally high insider buying and decreases to 0.612 (0.585) for firms where the accruals are accompanied by abnormal insider selling. These results are consistent with our proposition that insider trading information is useful in assessing the quality of the non-cash components of earnings. Fourth, we find that investors price all positive accruals as if they were informative. That is, we find that (1) investors correctly price positive accruals that are likely to be informative because concurrently insiders engage in abnormal buying and (2) investors price all positive accruals the same independent of insider trading. Thus, the mispricing of positive accruals is not pervasive. Absent insider buying, we find that the extent of the mispricing is far greater when positive accruals occur concurrently with abnormal selling than in cases where there is no trading. The extent of the mispricing when there is abnormal selling suggests that these accruals arise from opportunistic earnings management. By contrast, the smaller positive accrual mispricing when there is no insider trading is more likely related to the complexity of the firms' accrual generating process (Thomas and Zhang (2001) or to earnings fixation (Sloan 1996). Our evidence thus suggests that opportunistic earnings management is a partial explanation for the accrual mispricing phenomenon. Finally, we create implementable zero investment strategies that take short positions in firms with the highest accruals and abnormal insider selling and two alternatives for the

long position: (1) similar-sized firms or (2) firms with the highest cash flows.⁵ Our strategy which invests the proceeds from the short side of the hedge in similar-sized firms earns economically significant average one-year ahead returns ranging from 12.9% to 14.7% in the subsequent year and those where our long position is in high cash flow firms earns average returns ranging from 20.4% to 22.1%. By comparison, the corresponding hedge strategies which do not use insider trading yield one-year ahead returns of approximately 6 and 12 percent respectively, and a trading strategy based on insider trading activity alone yields a one-year ahead return of only 2.8%. Finally, we evaluate whether the returns to our hedge strategy can be explained by the fact that our ranking results in taking positions in firms that have other characteristics that are correlated with subsequent returns. We find that the returns to our hedge strategy are not simply manifestations of price to earnings, price to book, size, price momentum, or earnings surprise effects.

Our work makes three contributions to the existing literature examining accruals and earnings management. First, we provide evidence that an observable, contemporaneous signal, insider trading, is useful in evaluating whether accruals are more or less informative of future economic performance. This finding may be of particular interest to auditors who are now required to report to Audit Committees on the quality of a firm's earnings (Statement on Auditing Standards No. 89, December 1999). Second, while previous work documents that *on average*, the market misprices firms with large accruals, we find this result is not systematic, rather, we document that large positive accruals are rationally priced when they are accompanied by insider buying. Third, our finding that the greater mispricing occurs when positive accruals are accompanied by abnormal selling, suggests opportunistic earnings management. Given these results, we believe it is important to highlight the limitations of our study: We assume that our insider trading construct captures managers' private information and that our accrual metrics are reasonable proxies for earnings management. Failure in either assumption would introduce measurement error problems in our tests and we conduct several

⁵ Given our finding that negative accruals are not mispriced, we have no basis for taking a long position on firms with low accruals. Our evidence with respect to operating cash flows is similar to Sloan's and we take a long position on firms with high cash flows since their persistence appears under weighted by investors.

sensitivity tests in an effort to contain them.

The remainder of the paper is organized in four parts. Section 2 presents our empirical framework, Section 3 the data and methodology. The empirical results are presented in Section 4 and Section 5 concludes.

2. Empirical Framework

Assuming that earnings are managed, we view both managers' discretionary accounting choices and their trading behavior as signals of managers' private information. Our use of insider trading measures which can be positive (buying) or negative (selling) allow us to make directional predictions which are conditional on the sign of the accruals, therefore we partition the sample into positive and negative accruals. Our predictions relate to income increasing accruals for two reasons. First, reproducing Sloan (1996), we find that the accrual mispricing is due to positive accruals since negative accruals are rationally priced. Second, to generate joint insider trading and accrual predictions we rely on prior work on the effects of these signals on future stock prices. There is extensive evidence on the behavior of stock prices following buying and selling by insiders. However, evidence in previous work on the effect of earnings management on future prices is limited to income increasing accruals. To explain, there is evidence that firms likely to have inflated earnings are successful in *delaying* stock prices declines in Dechow et al. (1995, 1996) and Beneish (1997, 1999), but to our knowledge the effect, if any, of income decreasing earnings management on stock prices has not been documented.⁶

2.1 Hypotheses Development

If managers expect higher reported earnings to persist and lead to higher future stock prices, they have incentives to purchase their firm's stock. Such behavior is consistent with extensive findings from insider trading research that insiders buy in advance of stock price increases. Insider

⁶ Sloan (1996) cites a number of financial analysis texts proposing that *large positive* accruals are suggestive of earnings management. In particular he cites Bernstein (1993, p. 461) as stating that: "That is why analysts prefer to relate CFO to reported net income as a check on the quality of that income. Some analysts believe that the higher the ratio of CFO to net income, the higher the quality of that income. Put another way, a company with a high level of net income and a low cash flow may be using income recognition or expense criteria that are suspect."

buying may be a strong signal of earnings quality since open market purchases by insiders are far less frequent than sales due to the fact that they acquire most, if not all, of their shares through stock option and long term performance plans. Thus, if earnings management is intended to provide market participants with positive private information about their firm's prospects, we expect managers to either strategically increase their stake in the firms' equity (perhaps to provide another positive signal about firm prospects) or abstain from trading.⁷ Alternatively, if the earnings management is intended to hide deteriorating firm performance, for example, in an attempt to meet the market expectations of earnings or growth, we expect managers to act on their knowledge that the accruals are un-likely to persist and reduce their equity contingent wealth. Our expectation follows evidence in Beneish (1999) that managers overstate earnings to delay stock price declines and sell their equity contingent wealth at higher prices. Selling in the face of poorer firm prospects is also consistent with prior research findings that insiders sell in advance of price declines. If the earnings management is intended to mislead investors, managers could restrict their selling to reduce the likelihood of attracting the attention of the SEC's insider trading monitors. This would reduce the usefulness of the insider trading signal and bias against finding a relation between the interaction of insider trading, earnings management and future earnings. On the other hand, if the earnings management is successful in temporarily delaying declines in stock prices, increasing the time distance between the selling activity and the release of bad news reduces the likelihood that managers selling behavior is scrutinized by the SEC.⁸ A further reason to expect insider selling is provided by Beneish (1999) who shows that the SEC rarely imposes trading sanctions, even on insiders who sold

⁷ Managers may not acquire additional shares if they already own large equity stakes because additional equity can exacerbate their presumed under-diversification. That is, their human capital, salary, bonus, and equity contingent wealth (stock and options) are tied to firm performance.

⁸ The plaintiff's inability to establish scienter is often the stumbling block in numerous civil insider trading actions, where only preponderance of evidence is required. Scienter was defined by the Supreme Court under Rule 10b-5 as "a mental state embracing intent to deceive, manipulate or defraud." (*Ernst v. Hochfelder*, 425 U.S. 185, 193 (1976)) and must be proven to establish guilt in insider trading cases. Although insider trading in suspicious amounts or at suspicious times is probative of bad faith and scienter, nearly all the cases brought by the SEC between 1983 and 1996 are against corporate insiders, lawyers and brokerage house employees for using privileged information **immediately** preceding announcements of mergers, LBOs, and tender offers.

stock and were charged with fraudulently overstating earnings.⁹ The preceding discussion leads to the following hypotheses:

H₁: The persistence of income increasing accruals or unexpected accruals is lower when it occurs in combination with abnormal insider selling.

H₂: The persistence of income increasing accruals or unexpected accruals is higher when it occurs in combination with abnormal insider buying.

We evaluate these hypotheses by modifying the specification of the relation between current and future earnings in Sloan (1996) to accommodate interactions of accruals with our insider trading constructs and include the book to price ratio to control for the effect of growth on earnings. In addition, we evaluate the persistence of total and current accruals as well as their unexpected counterparts (see Appendix A) using several models based on Jones (1991).

Because we evaluate the hypotheses above with *raw* as well as unexpected accruals, we allow the predictions to obtain even in the absence of earnings management. To explain, managers' involvement in strategic operating decisions enables them to make better assessments of the likelihood that the accruals will result in future earnings. For example, a positive accrual that originates in an increase in receivables could be the result of an acceleration in sales that is suggestive of higher future sales growth or the result of operating decisions designed to avoid reporting lower sales growth, e.g., decisions to relax credit checks, grant easier credit terms, and/or front-load distributors.¹⁰ It is possible that an investor could distinguish between these two possibilities with a detailed financial analysis (e.g., tracking changes in gross margins) and/or by relying on non-financial information. However, Tarpley's (2000) evidence that accrual mispricing is

⁹ It is also possible that in the event of detection managers could justify their selling for personal liquidity reasons. For example, in *Provenz v. Miller* (1996) and in *Searls v. Glasser* (1995) appellate courts ruled that selling stock for the purpose of satisfying tax liabilities or purchasing a new or second home did not constitute illegal insider trading. Evidence that managers' compensation is reduced in times of financial distress (Gilson and Vestuypens (1993)) could also justify the sale of shares to maintain current consumption.

¹⁰ Similarly, a positive accrual that originates in an increase in inventory levels could be the result of an operating decision that anticipates higher future sales or the result of an unanticipated slowdown in demand.

most pronounced among young growth firms suggests the distinction is a difficult task, as young growth firms tend to have shorter, potentially less stable financial histories and poorer information environments (Barry and Brown (1984)). We expect that managers have information about the source of accruals and thus of the likelihood that the positive (negative) accrual will result in higher (lower) future earnings.

A priori, *raw* accruals are not estimates of earnings management. However, accruals and unexpected accruals are highly correlated: the Pearson correlations between raw total accruals and unexpected total accruals is 0.845, and the correlation between raw current accruals and unexpected current accruals measures is 0.873. These high correlation coefficients have two possible interpretations. First, if earnings have been managed and unexpected accruals are reasonable estimates of the earnings management, then so are raw accruals. This would occur, for example, if an adverse shock to firms' performance leads to earnings management and reduces the out-of-sample forecasting validity of Jones-like models of accrual expectations, thus making earnings management detectable by studying the raw accruals.

An alternative interpretation of the high correlations is one of no earnings management. In such case, the observed accruals and unexpected accruals are simply the results of economic changes in firm performance. To the extent that positive accruals results from a decline in firm performance, a finding that accruals are greater for abnormal selling firms suggests that managers sell following poor firm performance and in anticipation of price declines. Thus, the interpretation of our evidence depends on how one views management incentives to delay stock price declines.

The remaining hypotheses relate to the extent to which stock prices reflect the different properties of accruals and cash flows when accruals are conditioned on the extent of insider trading. As in Sloan (1996), we investigate whether investors "fixate" on earnings and fail to understand the differential persistence of accrual and cash flow components of earnings even after conditioning on insider trading signals. Because our goal is to evaluate whether investors are able to use insider trading information to make refined assessment of accrual persistence, our measures of insider

trading and accruals are *contemporaneous*.¹¹ We thus evaluate whether investors distinguish that accruals associated with insider buying (selling) are more (less) persistent than accruals associated with no insider trading. Our hypotheses patterned after Sloan's (1996, p. 292) are:

H₃: The earnings expectations embedded in stock prices fail to fully reflect the higher (lower) persistence of accruals or unexpected accruals associated with abnormal insider buying (selling).

H₄: Positive abnormal returns will be generated by a trading strategy taking a short position in firms with relatively high levels of accruals or unexpected accruals and abnormal insider selling and a long position in either: (1) a portfolio of similar-sized firms, or (2) firms with relatively high levels of cash flows.

To the extent that the earnings expectations embedded in stock prices fail to fully reflect the lower persistence of positive accruals associated with abnormal insider selling, the fourth hypothesis exploits these deviations from rational pricing on the short side of the hedge. The long side of the hedge is either a portfolio of similar-sized firms, or following Sloan (1996), firms with high cash flows. As with the first two hypotheses, we evaluate the persistence of current accruals as well as unexpected current accruals estimated using several models based on Jones (1991).

2.2 Relation Between Future Earnings and Components of Current Earnings

Our tests of the proposed hypotheses relating to the persistence of components of current earnings in future earnings are based on the approach in Sloan (1996). His tests use the following specification of the relation between current and future earnings (Sloan (p. 297)):

$$\text{Earnings}_{t+1} = a_0 + a_1 \text{CFO}_t + a_2 \text{Accruals}_t + \varepsilon_{t+1} \quad (1)$$

Earnings are measured as operating income after depreciation (Compustat #178). *Accruals* are calculated as the change in current assets (Compustat #4), minus the change in cash (#1) and current liabilities (#5), plus the change in the current portion of long term debt (#34) minus depreciation and

¹¹ We measure both earnings and insider trading over the identical January 1-December 31 period each year. Thus, the period in which earnings are generated is the same as the period over which insiders trade

amortization (#14) plus the change in taxes payable (#71). If items #34 or #71 are missing, then the item is set equal to zero in calculating total accruals since those numbers are likely immaterial for the firm in question. If any of the other required items are reported as missing in COMPUSTAT, the observation is dropped. All variables are deflated by average total assets (#6). Operating cash flow, *CFO*, is the difference between *Earnings* and *Accruals*.

Sloan (1996) provides evidence that the persistence of current earnings is decreasing in the magnitude of the accrual component of earnings and increasing in the magnitude of the cash flow component of earnings e.g. $a_1 > a_2$. Our modification of Sloan's estimation of equation (1) requires that we partition accruals (or unexpected accruals) into income increasing and income decreasing samples, and that we interact positive accruals with our insider trading measure. Thus, we conduct our tests using the following equation:

$$\begin{aligned} \text{Earnings}_{t+1} = & b_0 + b_1\text{CFO}_t + b_2\text{PosA}_t + b_3\text{PosA}_t*\text{SELL}_t + b_4\text{PosA}_t*\text{BUY}_t + b_5\text{NegA}_t \\ & + b_6\text{SELL}_t + b_7\text{BUY}_t + b_8\text{BP}_t + \varepsilon_{t+1} \end{aligned} \quad (2)$$

where,

CFO = Operating Cash Flow

PosA = accruals when accruals are positive and zero otherwise.

NegA = accruals when accruals are negative and zero otherwise.

SELL (*BUY*) are dummy variables indicating abnormal insider selling or buying activity.

BP = Book Value of Equity/ Market Value of Equity

Our measure of abnormal insider trading is described in section 3.2. We include book to price ratio in the model to control for growth following evidence that insider trading volume and proxies for growth are correlated [Seyhun (1986), Rozeff and Zaman (1988), Pope, et .al. (1990), Lin and Howe (1990)].¹² Our hypotheses predict that $b_3 < 0$ (H_1), $b_4 > 0$ (H_2). When examining current accruals we modify equation (2) slightly by adding depreciation as an explanatory variable. To analyze unexpected accruals, we employ several accrual estimation methods (described in Appendix A) and the following model specification:

$$\text{Earnings}_{t+1} = c_0 + c_1\text{CFO}_t + c_2\text{PosUA}_t + c_3\text{PosUA}_t*\text{SELL}_t + c_4\text{PosUA}_t*\text{BUY}_t + c_5\text{NegUA}_t + c_6\text{SELL}_t + c_7\text{BUY}_t + c_8\text{EACC}_t + c_9\text{BP}_t + \varepsilon_{t+1} \quad (3)$$

where, CFO, BP, BUY and SELL are defined as before, and
 PosUA = unexpected accruals when accruals are positive and zero otherwise.
 NegUA = unexpected accruals when accruals are negative and zero otherwise
 EACC = expected accruals

Using this model specification our hypothesis predict that $c_3 < 0$ (H_1), and $c_4 > 0$ (H_2).

2.3 Pricing Tests

Following Sloan (1996), we use the Mishkin (1983) test to examine whether the market rationally prices accruals and unexpected accruals conditional on insider trading. Subsequently, we employ hedge portfolio tests to ascertain the magnitude and economic significance of the returns associated with the mispricing.

The original application by Mishkin (1983) used a system of equations framework to test the rational expectations hypothesis in various macro-economics settings. We use the Mishkin test to examine whether the market rationally prices accruals given their implications for future earnings after we introduce insider trading information into the model. Specifically, we use the following regression system to implement the Mishkin test:

$$\text{Earnings}_{t+1} = a_0 + a_1\text{CFO}_t + a_2\text{PosA}_t + a_3(\text{PosA}_t*\text{SELL}_t) + a_4(\text{PosA}_t*\text{BUY}_t) + a_5\text{NegA}_t + a_6\text{SELL}_t + a_7\text{BUY}_t + a_8\text{BP}_t + v_{t+1} \quad (4a)$$

$$\text{SIZEAdJR}_{t+1} = \lambda + \beta[\text{Earnings}_{t+1} - \alpha_0 - \alpha_1\text{CFO}_t - \alpha_2\text{PosA}_t - \alpha_3(\text{PosA}_t*\text{SELL}_t) - \alpha_4(\text{PosA}_t*\text{BUY}_t) - \alpha_5\text{NegA}_t - \alpha_6\text{SELL}_t - \alpha_7\text{BUY}_t - \alpha_8\text{BP}_t] + \varepsilon_{t+1} \quad (4b)$$

where SIZEAdJR_{t+1} is the one-year ahead size-adjusted return starting four months after the fiscal year end in which accruals are measured, and all other variables are defined as before. Equation (4a) is a forecasting equation that estimates the coefficients of accruals and other earnings components

¹² Our insider trading metric measures abnormal insider trading after controlling for firm size and shares outstanding. The correlation between our Insider Trading measure and the Earnings to Price Ratio and Book to Price ratio were -0.007 and 0.065 respectively.

that best predict one year-ahead earnings, and equation (4b) is a valuation equation that estimates the valuation coefficients (α_i) that the market assigns to accruals, unexpected accruals and other earnings components. Following Mishkin we estimate equations (4a) and (4b) jointly using a two stage iterative generalized non-linear least squares procedure.

Initially the unconstrained model is estimated, that is, equations (4a) and (4b) are jointly estimated without imposing any constraints on the coefficients. In the second stage, the restricted model is estimated after imposing the rational pricing constraints ($\alpha_q = a_q$, for all q). To test the rational expectations hypothesis, i.e., whether the valuation coefficients and the forecasting coefficients obtained in the first stage are significantly different from each other a likelihood ratio is computed. Under the null hypothesis of rational expectations, Mishkin shows that the following test statistic is asymptotically $\chi^2(q)$ distributed:

$$2 \cdot N \cdot \log(SSR^c / SSR^u),$$

where q = the number of rational pricing constraints imposed,
 N = the number of observations in the sample
 SSR^c = the sum of squared residuals from the constrained regressions in the second stage,
 SSR^u = the sum of squared residuals from the unconstrained regressions in the first stage.

The rational pricing of any or all earnings components (i.e., $\alpha_q = a_q$) is rejected if the likelihood ratio is sufficiently large; i.e., if imposing one or all rational pricing constraints significantly reduces the model fit.

Our second set of pricing tests evaluate the returns to trading strategies that select firms based on their accruals and insider trading properties. Using Sloan (1996) as our benchmark we adopt a similar methodology and rank firms on the magnitude of their accruals and assign them in equal numbers to ten portfolios each year. In addition to total accruals, we examine strategies based on ranking firms according to the magnitude of their cash flow, current and unexpected current accruals. Our hypothesis predict that the mispricing will be greatest for those firms where insiders are abnormal sellers, therefore we partition the given decile into three mutually exclusive groups: abnormal selling, abnormal buying, and no insider trading firms. As in Sloan, size adjusted returns

are computed by taking the difference between the 12-month buy and hold return on a portfolio starting four months after the fiscal year end in which accruals are measured and the 12-month buy and hold return on a value weighted portfolio of firms of similar market value. We then investigate the returns to various zero investment strategies using firms from the highest accruals decile.

3.0 Method

3.1 Sample Selection

The initial sample consisted of all firms on the Compustat Industrial, Research, and Full Coverage 1998 tapes for the period 1985 to 1997. The sample was reduced by (1) eliminating financial services firms (SIC codes 6000-6700), (2) eliminating non-December fiscal year end firms, (3) firms with insufficient data to compute accruals, (4) firms with total assets less than one million dollars, and (5) firms with discontinued operations (Compustat #66) exceeding 5% of total assets. The rationale behind these criteria is as follows. As in much prior work, financial services firms are omitted because computing accruals is problematic. Only December fiscal year end firms are selected in order for the subsequent portfolio assignments to yield tradable investment strategies. In order to avoid extreme values caused by a small denominator, firms with average assets less than 1.0 (\$million) are omitted. Firms with significant reclassified operations are excluded since the prior year's financial data is not re-adjusted and the accrual computations rely on lagged financial data. The COMPUSTAT sample yielded 27,218 firm-year observations from 4770 different firms.

In order to perform stock returns tests, additional restrictions are placed on the sample used in the prior earnings based tests. Observations are deleted due to: (1) unavailability of 1999 return data on CRSP, thus 1997 data was removed since our return measurement period spanned the period from May 1998 to April 1999, (2) firms with missing return data on CRSP, yet data on COMPUSTAT, (3) firms that began trading after April in any given year are deleted from the sample in that year because these firms were not tradable at the time of portfolio formation, (4) firms with OMPUSTAT data for the prior year that are delisted prior to May portfolio formation date, and (5) American Depository Receipts are eliminated since CRSP does not compute a market capitalization size decile

for ADRs and thus size adjusted returns are not available. The resulting sample used in the stock return analysis consisted of 21,678 observations from 3906 firms over the period 1985 to 1996.

3.2 Insider Trading, Abnormal Buying and Abnormal Selling

Section 16(a) of The Securities Exchange Act of 1934 requires all insiders (officers, directors or 10 % beneficial owners) to disclose transactions in their firm's equity securities on SEC Form 4 on or before the 10th day of the month following the event. Thus, the prior year's insider trading data will be readily available at the time of accrual portfolio assignments (April). Insider trading data were obtained from the National Archives Center for Electronic Records and Thomson Financial. We use the same procedures and assumptions that have been used in the large body of prior research on insider trading, namely, we focus on open market (public) transactions and omit transactions of less than 100 shares. By convention, private transactions are omitted since it is assumed that reputation effects preclude insiders from opportunistic trading where both parties are known.

Prior insider trading research has documented the existence of an "information hierarchy" whereby higher-level insiders possess more valuable information and achieve larger abnormal returns when trading [Basel and Stein (1979), Nunn, et.al. (1983), Seyhun (1986, 1998), Lin and Howe (1990)]. We therefore construct our measure of insider trading using the transactions of the top five executives in the firm (CEO, CFO, COO, President, and Chairman of the Board), because they presumably possess the most valuable information. Descriptive statistics on the aggregate level of insider trading and two widely used insider trading measures are presented in Table 1. From Panel A, Table 1, both the frequency and volume of insider selling exceeds that of insider buying. The number of sales is twice the number of purchases while the number of shares sold (1.02 billion shares) outnumber purchases (0.35 billion shares) by almost 3:1. Following prior insider trading research, we compute an annual, firm specific, insider trading metric *Net Insider Trading Shares*.

$$NITS = \sum_i (\text{Shares Purchased}_{i,t} / \text{Shout}_t) - \sum_i (\text{Shares Sold}_{i,t} / \text{Shout}_t)$$

where the sum is over each firm's "i" senior executives over all days "t" in the calendar year, and, where *Shout* is the number of shares outstanding on the date of the insider transaction. We define net

shares traded in terms of the proportion of the firm traded to control for the cross-sectional variation in the absolute level of shares outstanding. From Panel B, we see that the number of net selling firms ($NITS < 0$) exceed the number of net buying firms ($NITS > 0$), while there is no insider trading activity by senior executives for approximately 70 percent of the firm-years. For subsequent sensitivity analysis, we also compute a second measure, *Insider Consensus*.¹³ Insider consensus, measures the number of unique insiders with purchase or sales transactions. In Panel B, we also observe that most firm's insiders' trading actions are unanimous, in only two percent of the sample does a senior insider sell shares in the same year that another senior insider purchases shares.

Table 2 examines the relation between insider trading, firm size and the magnitude of accruals. In Table 2, Panels A and B, we confirm prior findings that firm size and insider trading activity are strongly correlated; small firms have relatively more insider purchases than sales, while larger firms have relatively more insider sales than purchases [Seyhun (1986); Rozeff and Zaman (1988); Pope et al. (1990); and Lin and Howe (1990)]. In order to isolate “abnormal” insider trading, we first compute the annual NITS for net selling and net buying firms using all firms in the available CRSP population for each size decile. We then compute a size-adjusted insider trading measure for each sample observation by comparing its level of insider trading to the median level of all similar-sized firms in the same decile measured over the same year. Similar to the approach of Penman (1985), we then categorize the observation as:

Abnormal Selling = 1 if $NITS < 0$ and NITS is less than the median of all net selling firms
= 0 otherwise.

Abnormal Buying = 1 if $NITS > 0$ and NITS is greater than the median of all net buying firms
= 0 otherwise

In the extreme size deciles the relative number of purchases and sales is skewed and we modify our

¹³ Most insider trading research utilizes variations on the net shares and insider consensus metrics. Examples of the net shares metric include Penman(1982,1985), Elliot, et.al (1984), and Seyhun (1986). Insider consensus (i.e. the number of insiders trading) was prevalent in early research, e.g. Lorie and Neidhoffer (1968), Jaffe (1974), Finnerty (1976), and Madden (1979).

definition slightly and categorize all net selling firms in size decile 1 as abnormal selling firms. Similarly, we define all net buying firms in size decile 10 as abnormal buying firms. From Table 2, Panel C, the proportion of abnormal buying and selling observations is fairly uniform for accrual size deciles one to seven, while the three largest deciles are characterized by more abnormal selling and less abnormal buying firm-years.¹⁴

3.3 Sample Characteristics

We describe the characteristics of our sample in Tables 3 and 4. In Table 3, the statistics are organized by accrual deciles as in Sloan (1996) to allow comparisons across studies and because some of our subsequent return tests exploit such partitioning. Panel A describes the components of earnings and reveals, consistent with Sloan (1996), evidence of a negative correlation between accruals and cash flows (Pearson correlation -0.316, p-value <.0001). The mean (median) value of cash flow falls from 0.116 (0.178) in the lowest accrual decile to -0.106 (-0.045) in the highest accrual decile.¹⁵ Panel A also reveals evidence of a positive correlation between accruals and either unexpected total or unexpected current accruals for either the Jones (1991) or the Dechow et al. 1995 (hereafter DSS) modification of the Jones model. For example, for the DSS modification, the correlation between total accruals and unexpected total accruals is 0.845 (p-value <0.0001), and the correlation between total accruals and unexpected current accruals is 0.824 (p-value <0.0001).¹⁶ The mean (median) value of the DSS unexpected current accrual increases from -0.153 (-0.129) in the lowest accrual decile to 0.168 (0.144) in the highest accrual decile.

¹⁴ For our Insider Consensus measure we classify firm-years where there is unanimous selling (buying) as abnormal selling (buying). Note that insider consensus ignores the magnitude of shares traded, thus a firm year would not be considered an abnormal selling year if, for example, the CEO sold 20 million shares, while in the same year the CFO purchased 500 shares. We use Net Insider Trading as our primary measure following evidence from Seyhun (1986) that insider consensus (unanimity) is uniformly weaker than net share in explaining subsequent abnormal returns.

¹⁵ There are two noteworthy differences between our sample and Sloan's (1996). First, firms in our lowest accrual decile tend to have negative earnings (mean -0.133, median -0.044) where firms in the corresponding decile in Sloan's sample tend to be profitable with mean and median earnings of 0.07 (p. 296). Second, firms in our highest accrual decile tend to have lower earnings (mean 0.067) and cash flows (mean -0.106) than the firms in the corresponding decile in Sloan's sample (mean earnings 0.15, mean cash flows 0.00). Both differences suggest that firms in our extreme accrual deciles are less profitable than those in Sloan's sample.

¹⁶ The high correlations suggest that most of the variation in total accruals is accounted for by the variation in unexpected total accruals. We find similar results comparing current and unexpected current accruals ($p = 0.873$), suggesting that using accruals or unexpected accruals may lead to similar results for our earnings and returns tests.

In Panel B of Table 3, we present a number of firm characteristics to gain insights on the profile of firms that result from ranking on accruals. In addition, because our hedge portfolios are based on extreme deciles, the last column in this panel presents tests comparing the means and medians of the lowest and highest accrual decile. Relying on median tests, we find that firms in the lowest decile are smaller, more levered and less profitable. They also have lower returns in the prior year, lower earnings to price and book to price ratios. These observed differences lead us to evaluate whether the returns to our hedge strategy can be explained by the fact that our ranking results in taking positions in firms that have other characteristics that are correlated with subsequent returns.

In Table 4, we present the characteristics of our sample firms using a partition based on whether firms are abnormal sellers, abnormal buyers, and neither. We seek to identify whether there are differences between the three groups that could account for differences in the persistence of accruals. Identifying such variables would enable us to identify controls for our subsequent tests. Consistent with the results reported in Table 3, we find in Panel A that abnormal sellers (buyers) tend to have larger (smaller) earnings, accruals and unexpected accruals. Panel B reports comparison on a number of financial characteristics. Relying on median tests, abnormal sellers appear to be larger than abnormal buyers and similar in size to the Neither group; they are also less levered and more profitable, and have lower book-to-price ratios than the Abnormal buyers and Neither groups. In our subsequent hedge return tests we thus consider size, leverage, prior-year return and book-to-price as potential controls.

4. Empirical Results

4.1 Regressions of Future Earnings on Components of Current Earnings

In Table 5, we present our tests of H_1 and H_2 , by examining the persistence of accruals and unexpected accruals after conditioning on insider trading. Model (1) replicates Sloan (1996) with similar results: Our coefficient estimates of 0.860 and 0.752 for *Cash Flows* and *Accruals* respectively, are similar to his (0.855 and 0.765). As in Sloan, an F-test rejects the null hypothesis that the coefficients are equal at the 0.001 level and our results are thus consistent with the cash

component of earnings being more persistent than the accrual portion. Model (2) partitions accruals into positive and negative samples. While the persistence of positive and negative accruals are significantly less than that of cash flow, they are not significantly different from each other (F-test = 0.57, $p=0.56$).

Model (3) presents the results of our tests of persistence and insider trading. The interaction between abnormal selling and income increasing accruals is negative (-0.170) and significant at the 0.01 level. Thus, in support of H_1 , we find that the persistence of income increasing accruals is significantly lower when accompanied by abnormal insider selling. In contrast, the interaction between positive accruals and insider buying is positive (0.141) and significant at the 0.01 level. This result supports H_2 , that the persistence of income increasing accruals is significantly greater when we observe abnormal insider buying. In Model (4) we obtain virtually identical results after adding insider trading dummy variables to control for the effect of insider trading on earnings and the book to price ratio to control for growth. In Model (5) we examine current accruals and obtain similar results. We interpret the results of Table 5 as strong evidence in support of H_1 , and H_2 , that insider trading signals are informative about earnings quality.

We extend our analysis to unexpected accruals in Table 6. The first two columns examine the persistence of unexpected current accruals for the Jones (1991) and DSS modification of the Jones model. In both representations, we find that the persistence of income increasing unexpected current accruals is significantly lower in firms where there is abnormal selling. In contrast, abnormal buying is associated with significantly higher persistence. We obtain similar results for unexpected total accruals. If our unexpected accrual measures capture earnings management, the evidence suggests that insider trading is useful in distinguishing opportunistic from informative income increasing earnings management.

4.2 Returns Tests Results.

4.2.1 Mishkin Tests

In Table 7, Panel A, we reproduce Sloan (1996)'s pricing tests and obtain similar results. The market appears to misprice accruals and cash flows, overestimating the persistence of accruals while underestimating that of cash flow. In Panel B, we examine whether the observed mispricing is characteristic of both income increasing and income decreasing accruals. We find that the market misprices positive accruals; the estimate of persistence implicit in the pricing regression (1.011) is significantly greater than the persistence implied from prior earnings information (0.743). In contrast, we find no evidence that the market significantly misprices negative accruals, suggesting that the accrual mispricing phenomenon is largely driven by positive accruals. This finding, along with evidence in Tarpley (2000) that the mispricing is particularly pronounced among growth firms, leads us to propose that the accrual mispricing arises because it is inherently difficult to: (1) distinguish whether positive accruals signal growth acceleration or deceleration, or (2) see through income increasing earnings management.

In Panel C, we investigate whether insider trading is useful in explaining the mispricing of positive accruals. The interaction between positive accruals and abnormal insider selling in the forecasting equation is -0.171 suggesting lower persistence, but the estimate from the pricing regression, 0.235, suggests investors attach higher persistence to positive accruals accompanied by abnormal insider selling. Indeed, adding the main effect (coefficients a_2 and α_2) to these interaction effects we find that, in cases of abnormal insider selling, the overall persistence of positive accruals implicit in the pricing equation (1.196) is more than double that obtained from the earnings regression (0.590). This difference is significant, the likelihood ratio for the hypothesis that $\alpha_2 + \alpha_3 = a_2 + a_3$ is 12.44 (p-value <0.001). When there is no insider trading, positive accruals are also mispriced, although the effect is less pronounced. In these cases, the estimate of persistence obtained from the pricing equation (0.961) is significantly greater than the forecasting equation estimate (0.761). Examining abnormal insider buying, we find that there is no significant mispricing of

accruals: The incremental persistence of positive accruals occurring with insider buying is 0.140 in the earnings regression, while the estimate obtained from the pricing regression is 0.056. Adding the main and interactive effects, the persistence of positive accruals from the pricing equation (1.017) is not significantly different from that implied from the earnings regression (0.901) (p-value =0.657).

In Panel D, we find that the mispricing results we observe for total accruals are also present for current accruals. Income-increasing current accruals are priced rationally in firms where they are accompanied by abnormal buying, and mispriced when concurrently there is abnormal selling, and mispriced to a lesser degree when there is no insider trading.¹⁷ In Panel E, we find similar results when we examine unexpected current accruals based on the DSS model.

When we include insider trading signals in our analysis we consistently find that the extent of the accrual mispricing is greater when positive accruals occur concurrently with abnormal selling than in cases where they do not. For example, from Panel C, when there is abnormal selling, the market overestimates the persistence of positive accruals by 103 percent (1.196/0.590) whereas the market only overestimates their persistence by 26 percent (0.961/0.761) when there is no insider trading. If the mispricing arises because participants do not fully understanding the nature of firms' accrual generating process (such as in new or high growth firms), we would not expect accrual mispricing to be related to insider trading. On the other hand, if the positive accruals that occur concurrently with abnormal selling are the result of opportunistic earnings management, we would expect to subsequently observe lower earnings persistence and greater relative mispricing.

We perform several tests to examine the sensitivity of our findings to our proxies for abnormal insider trading and unexpected accruals. In Table 8, Panels A and B, we re-examine whether our results are sensitive to the choice of insider trading metric. Employing the alternative measure, *Insider Consensus*, we obtain qualitatively similar results and our conclusions remain

¹⁷ When there is abnormal insider selling, the persistence estimated from the pricing equation is 1.194 while the persistence of current accruals implied from forecasting equation is 0.609. This difference is significant, we obtain a likelihood ratio test statistic of 13.61 (p-level <0.001) leading us to reject the hypothesis that $\alpha_2 + \alpha_3 = \alpha_2 + \alpha_3$. We fail to reject that positive current accruals are mispriced when there is insider buying, the likelihood ratio test statistic is 0.33 (p-level 0.561) for the hypothesis that $\alpha_2 + \alpha_4 = \alpha_2 + \alpha_4$.

unchanged. We also replicate our analysis of Table 7 (not presented) using the Jones model of expected accruals. In all cases our conclusions remain unchanged.

In summary, our earnings tests suggest that insider trading signals are useful in evaluating the earnings quality of income increasing accruals, i.e., the extent to which they are predictive of future earnings. However, our evidence indicates that the market rationally prices only those accruals accompanied by abnormal buying, and, hence where the accrual's persistence is greatest. The mispricing of positive accruals and unexpected accruals when there is abnormal selling is attributable to the market overestimating their persistence. One possible explanation for our findings is that the market prices all accruals as if they are as informative as those accompanied by abnormal buying. Examining the market pricing equation for positive current accruals (Table 7, Panel D), we obtain estimates of 1.194, 0.920, and 0.980 when there is abnormal selling, no insider activity, and abnormal buying respectively. We cannot reject the hypothesis of joint equality for the three parameters: Testing the restriction that $\alpha_2 + \alpha_3 = \alpha_2 + \alpha_4 = \alpha_2$, we obtain a likelihood ratio of 2.79 (p-level 0.248). We find similar results when we examine unexpected current accruals. We interpret these results as consistent with the market pricing all accruals as if they were informative. However, this result also leads to the question as to why the market apparently does not efficiently utilize insider selling signals. Our measure of insider trading is adjusted for firms' market capitalization and normalized by the shares outstanding on the transaction date and is based on a subset of insiders of the firm. However, market participants may have difficulty distinguishing insider selling that conveys information about the future from insider selling triggered by non-information events such as managers' liquidity needs, portfolio rebalancing needs, and tax-related events.

We next examine whether we can profitably exploit our findings and, to allow for a comparison to Sloan (1996)'s result, we evaluate several strategies based on insider trading signals and extreme accrual and cash flow deciles.

4.2.2. Portfolio Tests

In Table 9, we examine the returns to various portfolios comprised of firms based on their accrual and insider trading characteristics. As in Sloan (1996) we compute in Panel A mean size-adjusted returns for deciles formed on total accruals. The return to the hedge portfolio formed by taking a long position in the lowest accrual decile and a short position in the highest accrual decile earns a hedge return of 11.0 percent, which is similar to Sloan's reported 10.4 percent (1996, p. 307). The slight difference is due to the long side of the hedge portfolio as our return on the short side of the hedge, -5.5 percent, exactly matches Sloan's result.

Incorporating insider trading information in the hedge portfolio formation leads to higher hedge returns. A hedge return of 14.7 percent can be obtained by shorting abnormal selling firms in the highest accrual decile, and investing the proceeds in a portfolio of similar-sized firms. Through the use of our insider trading signal, we are able to identify firms that are subsequently poor performers. The average return of -5.5 percent for the high accrual decile can be decomposed into -14.7 percent in firms where insiders are abnormal sellers, -2.5 percent in firms where there is abnormal buying, and -3.9 percent in firms where there is no trading. We consider the possibility that our insider trading measure is simply identifying those firms with the most extreme accruals in the high total accrual decile. We find that the mean and median accrual levels for the abnormal selling and non-abnormal selling groups are not statistically different.¹⁸

Given our finding that negative accruals are not mispriced, we have no basis for taking a long position on firms with low accruals. Our evidence with respect to operating cash flows is similar to Sloan's and we take a long position on firms with high cash flows since their persistence appears under-weighted by investors. The mean size adjusted return for firms in the highest cash flow decile was 7.4 percent ($t=5.72$). Thus, for this hedge portfolio we obtain an overall return of 22.1 percent

¹⁸ For decile 10, the mean total accrual was 0.168 for the abnormal selling sample, and 0.174 for the sample complement ($t=0.92$, Wilcoxon $z=0.88$). The mean current accrual was 0.208 for the abnormal selling sample and 0.214 for the sample complement ($t=0.89$, Wilcoxon $z=0.61$). Similar results were obtained using the various unexpected accrual models.

(14.7 percent + 7.4 percent). Results for current and unexpected current accruals in Panels B and C are similar.¹⁹

We perform additional analysis to evaluate the sensitivity of our results. The hedge returns for unexpected accruals are numerically but not statistically different than those based on total or current accruals suggesting that the results are independent of whether or not we use an accrual expectation model. This result is not surprising because our accrual and unexpected accrual proxies are highly correlated ($\rho \approx 0.85$). If, however, the accrual expectation models do measure, albeit with error, managerial discretion, we should observe a different pattern for expected relative to unexpected accruals. We thus construct hedge portfolios based upon *expected* accruals. We find that hedge returns are not significant for portfolios based on expected current accruals.

In summary, the use of insider trading information in conjunction with accruals, yields hedge returns that are significantly higher than those obtained using accruals alone. The ability of the insider trading information to identify low earnings quality firms in the high accrual decile is robust across various accrual and unexpected accrual models. Furthermore, confirming our conjecture that the market mispricing is attributable to the market pricing all firms as if their accruals were informative, the average return for the high accrual decile-insider buying firms is insignificant in all three accrual models.

4.3 Alternative Explanations for the Observed Returns

In this section, we evaluate whether the returns to our hedge strategy can be explained by the fact that our ranking results in taking positions in firms that have other characteristics that are correlated with subsequent returns. Using the approach in Holthausen and Larcker (1992), we consider five variables (i) the earnings to price ratio, (ii) the book to price ratio, (iii) returns in the prior year, (iv) firm size, and, (v) earnings surprise.

¹⁹ We examined the annual returns to the various hedge portfolios we consider on a year by year basis. The annual hedge returns based on accruals are positive in 10 of 12 years. This result is similar to Sloan's (1996) findings and suggests that the sign of the hedge return appears stable over time. The annual hedge returns to a strategy that combines insider trading and current or unexpected current accruals is positive for 11 of 12 years. Hedge returns utilizing insider trading information yield higher returns than the accrual only counterpart in 10 of 12 years.

Table 10 reports the results of seven different regressions. Models (1) to (3) relate to hedge strategies based on accruals only, model (4) utilizes only insider trading measures in forming the hedge portfolio, and regressions (5) to (7) analyze hedge portfolios formed using accrual, cash flow, and insider trading measures. The dependent variable in each regression is the one-year ahead size-adjusted return (EXRET) to each of the firms in our hedge portfolios. The indicator variable HEDGE takes the value of 1 when we took a long position in the firms' stock and zero otherwise. The control variables considered are: (i) the earnings to price ratio (E/P), following evidence that firms with low P/E ratios outperform firms with high P/E ratios on a risk-adjusted basis (Fluck et al. (1993), Haugen and Baker (1996)); (ii) the book to price ratio (BP), following evidence in Chan et al. (1996), Davis (1994), Haugen and Baker (1996) who document that firms with high BP subsequently earn higher returns [BP is computed as the ratio of common stockholders' equity (Compustat #60) to market value of equity at the fiscal year-end immediately before the year in which the hedge portfolio is formed], (iii) returns in the prior year (RET_{t-1}) following evidence of price momentum in Jegadeesh (1990), Jegadeesh and Titman (1993) [RET_{t-1} computed as the size-adjusted return for the fiscal year preceding the year in which the hedge portfolio is formed], (iv) firm size (LMVE) following evidence in among others, Fama and French (1992), that size explains future returns [LMVE is computed as natural logarithm of the market value of equity at the fiscal year-end immediately before the year in which the hedge portfolio is formed], (v) and earnings surprise (SURP) to capture the post-announcement drift documented among others by Bernard and Thomas (1989) [SURP is computed as the difference between income before extraordinary items (Compustat Quarterly #8) in the fourth quarter in the most recent year and the corresponding quarterly earnings in the previous year, deflated by market value of equity at the end of the most recent fiscal year]. We include in the regression all but the earnings to price ratio, because the set of explanators already includes earnings or an earnings decomposition. The regression specification is:

$$EXRET_{it} = a_0 + a_1HEDGE_{it} + a_2BP_{it} + a_3RET_{it-1} + a_4EPRATIO_{it} + a_5LMVE_{it} + a_6SURP_{it} \quad (5)$$

All independent variables except the HEDGE dummy variable are mean adjusted. As a result, the slope coefficient on HEDGE, a_1 , can be interpreted as the return to the hedge portfolio after controlling for the effect of the other variables. The intercept (a_0) can be interpreted as the return to the short position in the hedge portfolio and the sum of a_0 and a_1 can be interpreted as the return to the long position in the portfolio after controlling for the effect of the other variables.

Equation (5) is estimated in a pooled cross-sectional regression as well as in year-by-year partitions. Because we obtain very similar results in both analysis, we present and discuss only the pooled results. The regressions using returns from the accruals-only hedge portfolios as dependent variables (columns 1 and 2) indicate returns on the strategies of 10.9 percent (total accruals) and 11.9 percent (current accruals), each significant at the 0.001 level after controlling for the effects of other factors. The hedge return of 11.8 percent for unexpected current accruals is nearly identical to that obtained from the raw current accruals. In all cases, the results are reasonably close to those reported in Table 9 when other factors that may account for the strategy's returns are not present.

A strategy based solely on insider trading yields a hedge return of only 2.8 percent (Table 10, Model 4). This result is consistent with size adjusted returns reported by Rozeff and Zaman (1988), Lin and Howe (1990), and Pope, et.al (1990). In models (5) through (7) we introduce insider trading and examine the robustness of our hedge returns due to strategies based on taking short positions in firms which exhibit large accruals and abnormal insider selling. Model (5) examines the most conservative of these strategies and finds a return of 12.9 percent when we invest the proceeds of the short size of the hedge in firms of similar size. In models (6) and (7) we capitalize on our findings that cash flows are under weighted and invest the proceeds from the short size of the hedge in firms with high cash flows. We obtain similar hedge returns of approximately 22 percent for the current and unexpected current accruals based hedges.

Consistent with the results of Table 9, employing accrual estimation models to compute unexpected accruals does not significantly improve hedge returns relative to strategies based on either total or current accruals. After controlling for possible determinants of future returns, the

hedge returns are nearly identical to those reported in Table 9, thus, using insider trading information improves our returns by approximately 10 percentage points.

Overall, we find that the profitability of our hedge strategy is not diminished by including the control variables, and the evidence suggests that the returns to our hedge strategy are not simply manifestations of P/E, P/B, size, price momentum, or post-earnings surprises effects.

4.4 Additional Analyses

We performed additional analyses to examine the sensitivity and validity of our results. If the returns to our trading strategy are due to our superior ability to isolate firms where earnings are misleading and thus where we expect poor stock performance, then the potential returns to our hedge should diminish once the subsequent earnings fail to materialize. Thus, we computed the returns to our hedge for the second year after portfolio formation. Returns for the various accrual and unexpected accrual hedges are statistically insignificant, averaging approximately 0.5 percent. Returns for the combination accrual and insider trading hedges are approximately 5 percent, achieve only marginal statistical significance (p-level ~ 0.10) and are far smaller than the 20 percent returns in the prior year. We interpret these results as being consistent with the market becoming informed, and our trading profits diminished when subsequent earnings realizations reveal a lack of earnings persistence

5. Conclusion

The paper provides evidence that the signal contained in insiders' trading behavior is useful in making refined assessments of earnings quality, and informative about the valuation implications of accruals. We find that accruals and unexpected accruals have lower (higher) persistence when managers are abnormal sellers (buyers). The result suggests that insider trading information is useful in assessing the quality of the non-cash components of earnings and can be useful to investors and especially auditors who are now required to report to Audit Committees on their clients' quality of earnings.

We show that the accrual mispricing phenomenon observed in previous research is largely due to the mispricing of positive accruals. We find that investors price all positive accruals as if they were informative. That is, we find that: (1) investors correctly price positive accruals that are likely to be informative because concurrently insiders engage in abnormal buying and (2) investors price all positive accruals similarly, independent of insider trading. Thus, the mispricing of positive accruals is not pervasive. Absent insider buying, we find that the extent of the mispricing is far greater when positive accruals occur concurrently with abnormal selling than in cases where there is no trading. The extent of the mispricing when there is abnormal selling suggests that these accruals arise from opportunistic earnings management. By contrast, the smaller positive accrual mispricing when there is no insider trading is more likely related to the complexity of the firms' accrual generating process [(Thomas and Zhang (2001))] or to earnings fixation [Sloan (1996)]. Our evidence thus suggests that opportunistic earnings management is a partial explanation for the accrual mispricing phenomenon.

We further validate our results by forming hedge portfolios which exploit the large mispricing of opportunistic accruals. We show that tradable strategies based on income-increasing accruals and insider trading activity earn economically significant one-year ahead returns of approximately 22 percent and that these returns dominate strategies based on either accruals (11 percent) or insider trading (2.8 percent) individually. Consistent with our prior findings that positive accruals and insider buying are priced correctly, we find that the average returns to a portfolio of these firms are insignificant. Consistent with the market becoming informed regarding the transitory nature (low persistence) of opportunistic accruals we find that hedge returns in the 2nd year after formation are also insignificant. Finally, we evaluate whether the returns to our hedge strategy can be explained by the fact that our ranking results in taking positions in firms that have other characteristics that are correlated with subsequent returns and find that the returns to our hedge strategy are not simply manifestations of P/E, P/B, size, price momentum, or earnings surprise effects.

Interpreting the results of our study requires assuming that: (1) our accrual metrics are appropriate surrogates for managers' accounting manipulations, and (2) managers *trade*

opportunistically based on their knowledge of the persistence of accruals (whether they have managed accruals or not) and our insider trading constructs capture managers' private information about their firms' accruals. Given these assumptions, our work makes three contributions to the existing literature examining accruals and earnings management. First, we provide evidence that an observable, contemporaneous signal, insider trading, is useful in evaluating whether accruals are more or less informative of future economic performance. Second, while previous work documents that *on average*, the market misprices firms with large accruals, we find this result is not systematic, rather, we document that large positive accruals are rationally priced when they are accompanied by insider buying. Third, our finding that the greater mispricing occurs when positive accruals are accompanied by abnormal selling, suggests that opportunistic earnings management is a partial explanation for the accrual mispricing phenomenon.

Appendix A

Accrual Expectation Models

We use three accrual expectation models derived from the Jones (1991) model. We estimate both unexpected *total accruals*, and unexpected *current accruals* models following DeFond and Jambalvo (1994) and Beneish (1998). The model proposed by Jones follows Kaplan's (1985) suggestion that accruals likely result from the exercise of managerial discretion and from changes in the firm's economic conditions. The model relates total accruals (Accruals) to the change in sales (ΔSales) and the level of gross property, plant and equipment (PPE):

$$\text{Accruals}_{it} = a_{1i} + b_{1i} \Delta\text{Sales}_{it} + c_{1i} \text{PPE}_{it} + u_{1it} . \quad (\text{A.1})$$

The model is based on two assumptions. First, that working capital accruals resulting from changes in the firm's economic environment are related to changes in sales, or sales growth since equation (A.1) is typically estimated with all variables deflated by either lagged assets or lagged sales. Second, that gross property plant and equipment controls for the portion of total accruals related to expected depreciation expenses. The second version uses current accruals (CACC) as a dependent variable and only the change in sales as an explanatory variable:

$$\text{CACC}_{it} = a_i + b_{2i} \Delta\text{Sales}_{it} + u_{2it} . \quad (\text{A.2})$$

We also consider two modified versions of the Jones (1991) model. In their total accrual form, the models are given by:

$$\text{Accruals}_{it} = a_{3i} + b_{3i} (\Delta\text{Sales}_{it} - \text{Receivables}_{it}) + c_{3i} \text{PPE}_{it} + u_{3it} . \quad (\text{A.3})$$

$$\text{Accruals}_{it} = a_{4i} + b_{4i} \Delta\text{Cash Sales}_{it} + c_{4i} \text{PPE}_{it} + u_{4it} \quad (\text{A.4})$$

The expectation model in equation (A.3) is typically attributed to Dechow et al. (1995) (e.g., see Gaver et al. (1995)), even though, as argued in Beneish (1998), the modified-Jones model presented in Dechow et al. (1995) is the same as the Jones model in the estimation period and only has the receivable adjustment in the prediction period. Indeed, the revenue based variable in (A.3) equals $\text{Cash Sales}_{it} - \text{Sales}_{it-1}$. Because it is not clear what the construct means or how it proxies for the effect on accruals of changes in the firm's economic environment, Beneish (1998) proposed an alternative modification based on cash sales (equation A.4). The evidence indicates that change in cash sales preserves the intent of using cash sales and has the advantage of using as an explanator an accounting construct that reduces the endogeneity problem.²⁰

The paper estimates these models cross-sectionally using all firms in a given two-digit industry and year (cross-sectional estimation has the advantage of not restricting the analysis to firms with long time-series of data). Each yearly estimation is used to make one-year ahead forecasts of expected accruals which, subtracted from the dependent variable, yields unexpected accruals.

²⁰ To explain; it is much harder to exercise discretion over cash sales than over credit sales. Indeed, examining firms whose financial reporting behavior is deviant enough to warrant SEC enforcement actions, Beneish (1997) finds that cash sales are rarely manipulated. One firm out of 64 (1.6%) engages in circular transfers of money to create the impression of receivable collection. In contrast, 43 of 64 firms (67.2%) engage in manipulations affecting credit sales (e.g., fictitious invoices, front loading with a right of return, keeping books open past the end of the fiscal period, overstating the percentage of completion).

Our study assumes that accruals and unexpected accruals are reasonable proxies for earnings management. Recent work that has studied the properties of unexpected or unexpected accruals generated with models based on Jones (1991) and concluded that such proxies measure earnings management with error. Our attempts to contain the measurement error problem include (1) using total and current accruals, (2) considering several models to partition accruals, and (3) assessing whether the same results obtain for expected vs. unexpected accruals.

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Table 1
Insider Trading Statistics and Measures
Senior Executives only, 1985-1996

Panel A: Aggregate Insider Trading Statistics		
Number of Firms	3906	
Number of Unique Senior Executive	6735	
Insiders with one or more transactions		
	<u>Purchases</u>	<u>Sales</u>
Open Market Transactions	9376	20792
Firm-years	2940	4162
Shares traded	352.6 M	1020.2 M
Panel B: Insider Trading Metrics		
<i>1. Net Shares</i> ^(a)	<u>Firm Years</u>	<u>Percent of Sample</u>
Net Selling Firms	4020	18.5 %
Net Buying Firms	2617	12.1
No Trading	15041	69.4
<i>2. Insider Consensus</i> ^(b)		
Unanimous Selling	3711	17.1
Majority Selling, not unanimous	50	0.2
Equal (nonzero) sellers and buyers.	366	1.7
Unanimous Buying	2489	11.5
Majority Buying, not unanimous	35	0.2
No Trading	15027	69.3
All transactions are open market purchases and sales coded "P" and "S" on SEC Form 4. Transactions of less than 100 shares, warrants, and convertible securities are omitted. Senior Executives are limited to the following positions: CEO, CFO, COO, President, and Chairman of the Board.		
(a) - Insider trading for each firm is measured as $(\text{Shares Purchased}_{i,t} / \text{Shout}_t) - (\text{Shares Sold}_{i,t} / \text{Shout}_t)$ where shares are purchased or sold by executive i on day t . Shout is the number of shares outstanding on the date of the insider transaction.		
(b) - This measures the number of insiders purchasing shares relative to the number of insiders selling shares. Unanimous selling occurs when one or more insiders of the firm sell shares while no insiders purchase shares in a given year. An example of Majority selling would be if two insiders sold shares during the year while one insider bought shares. The difference in the number of "No trades" is due to 14 occurrences of insiders omitting the number of shares in their filing.		

Table 2
The Relationship between Insider Trading, Firm Size, and Accruals
 Firms Assigned to Deciles Annually
Sample Consists of 21678 Firm-years between 1985 and 1996.

Panel A: Insider Buying and Selling versus Firm Size ^(a)										
Market Capitalization	Lowest Decile	2	3	4	5	6	7	8	9	Highest Decile
Net Insider Selling Firms	31.3 %	35.4 %	40.8 %	42.4 %	46.9 %	51.7 %	55.5 %	60.4 %	64.5 %	73.8 %
Net Insider Buying Firms	68.7 %	64.6 %	59.2 %	57.6 %	53.1 %	48.3 %	44.5 %	39.7 %	35.4 %	26.2 %
Panel B: Net Insider Trading Shares and Firm Size ^(b)										
Market Capitalization	Lowest Decile	2	3	4	5	6	7	8	9	Highest Decile
Median	0.170 %	0.069 %	0.033 %	0.014 %	-0.009 %	-0.052 %	-0.060 %	-0.073 %	-0.063 %	-0.020 %
Net Selling Firm Median	-0.846 %	-0.675 %	-0.701 %	-0.627 %	-0.551 %	-0.530 %	-0.498 %	-0.375 %	-0.226 %	-0.047 %
Net Buying Firm Median	0.442 %	0.219 %	0.190 %	0.119 %	0.084 %	0.070 %	0.046 %	0.009 %	0.016 %	0.005 %
Panel C: Abnormal Insider Trading and Accrual Deciles ^(c)										
Total Accruals	Lowest Accruals	2	3	4	5	6	7	8	9	Highest Accruals
Proportion of Abnormal Selling Firms	7.63 %	7.39 %	7.94 %	7.14 %	7.73 %	5.78 %	7.27 %	10.45 %	12.82 %	15.59 %
Proportion of Abnormal Buying Firms	8.02 %	8.08 %	7.57 %	7.05 %	8.01 %	8.12 %	7.36 %	6.86 %	6.43 %	5.71 %

Net Insider Trading Shares (*NITS*) for each firm is measured as $(\text{Shares Purchased}_{i,t} / \text{Shout}_t) - (\text{Shares Sold}_{i,t} / \text{Shout}_t)$ where shares are purchased or sold by executive *i* on day *t*, and normalized by the number of shares outstanding on the transaction date. Net selling (buying) firms are those where *NITS* < 0 (*NITS*>0).

(a) - Percentages are based on firms which report at least one insider transaction.

(b) - The reported medians are the average of the annual medians computed for years 1985-1996.

(c) - Abnormal insider trading is measured relative to the level of insider trading in the same year by firms of equal size (market capitalization decile).
 Abnormal Selling =1 if *NITS*<0 and *NITS* is less than the median of all net selling firms of similar market value, and zero otherwise.
 Abnormal Buying = 1 if *NITS*>0 and *NITS* is greater than the median of all net buying firms of similar market value, and zero otherwise.

Table 3
Mean (Median) Values of Selected Characteristics for Ten Portfolios of Firms Formed Annually by
Assigning Firms To Deciles Based on the Magnitude of Total Accruals.
Sample Consists of 21678 Firm-years between 1985 and 1996.

Panel A: Component of Earnings										
	Lowest Accruals	2	3	4	5	6	7	8	9	Highest Accruals
Total Accruals	-.250 (-.206)	-.118 (-.116)	-.084 (-.084)	-.063 (-.064)	-.047 (-.047)	-.033 (-.033)	-.017 (-.017)	.005 (.005)	.042 (.041)	.173 (.136)
Current Accruals	-.162 (-.132)	-.043 (-.045)	-.020 (-.022)	-.008 (-.008)	.002 (-.001)	.010 (.007)	.023 (.020)	.043 (.041)	.082 (.080)	.209 (.174)
Cash Flows	.116 (.178)	.125 (.165)	.126 (.154)	.121 (.143)	.110 (.131)	.097 (.118)	.080 (.105)	.057 (.083)	.022 (.051)	-.106 (-.045)
Earnings	-.133 (-.044)	.007 (.047)	.042 (.070)	.058 (.080)	.063 (.086)	.064 (.085)	.063 (.086)	.062 (.085)	.064 (.092)	.067 (.104)
Jones Disc. Total Accruals	-.176 (-.150)	-.059 (-.060)	-.031 (-.034)	.018 (-.018)	-.006 (-.007)	.001 (.004)	.011 (.011)	.024 (.024)	.050 (.051)	.141 (.121)
DSS Disc. Total Accruals	-.187 (-.158)	-.062 (-.064)	-.033 (-.035)	-.019 (-.019)	-.007 (-.008)	.001 (.003)	.012 (.012)	.028 (.027)	.057 (.057)	.162 (.139)
Jones Disc. Current Accruals	-.144 (-.120)	-.045 (-.043)	-.024 (-.023)	-.015 (-.012)	-.005 (-.004)	.000 (.002)	.010 (.010)	.025 (.026)	.053 (.054)	.144 (.124)
DSS Disc. Current Accruals	-.153 (-.129)	-.046 (-.044)	-.023 (-.022)	-.013 (-.011)	-.003 (-.003)	.003 (.003)	.014 (.013)	.031 (.031)	.063 (.064)	.168 (.144)

Total accruals = are calculated as the change in current assets (COMPUSTAT item #4), minus the change in cash (#1), minus changes in current liabilities (#5), plus the change in the current portion of long term debt (#34) minus depreciation and amortization expense (#14) plus change in taxes payable (#71). If items #34 or #71 are missing, then the item is set equal to zero in calculating total accruals since those numbers are likely immaterial for the firm in question. If any of the other required items is reported as missing in COMPUSTAT, that observation is dropped.

Cash Flow from Operations = (Operating Income after depreciation (#178) - Accruals)/Average total assets(#6).

Earnings = Operating Income after depreciation (#178)/Average total assets(#6)

Unexpected total accruals are estimated from two expectations models:

(1) Jones' (1991) Model: The model relates total accruals (TACC) to the change in sales (ΔSales) and the level of gross property, plant and equipment (PPE #8).

$$\text{TACC}_{it} = \alpha_{1i} + \beta_{1i} \Delta\text{Sales}_{it} + \gamma_{1i} \text{PPE}_{it} + \varepsilon_{1it}$$

(2) DSS modification: The Jones' (1991) model as modified by (Dechow Sloan and Sweeney 1995, [DSS]) relates total accruals (TACC) to the change in sales less changes in receivables (#2) ($\Delta\text{Sales} - \Delta\text{Receivables}$) and the level of gross property, plant and equipment (PPE #8)

$$\text{TACC}_{it} = \alpha_{1i} + \beta_{1i} (\Delta\text{Sales}_{it} - \Delta\text{Receivables}_{it}) + \gamma_{1i} \text{PPE}_{it} + \varepsilon_{1it}$$

Unexpected current accruals are estimated from similar expectations models. That is, the dependent variable is current accruals (CACC) which equals TACC + Depreciation, and the models have just one explanatory variable since the PPE variable is dropped.

Table 3-Continued
Mean (Median) Values of Selected Characteristics for Ten Portfolios of Firms Formed Annually by
Assigning Firms To Deciles Based on the Magnitude of Total Accruals
Sample Consists of 21678 Firm-years between 1985 and 1996.

Panel B: Financial and Market Characteristics											
	Lowest Accruals	2	3	4	5	6	7	8	9	Highest Accruals	Mean Test (Median Test)
Size	3.260 (3.095)	4.413 (4.257)	5.068 (4.912)	5.389 (5.431)	5.577 (5.675)	5.550 (5.662)	5.196 (5.132)	4.831 (4.781)	4.399 (4.347)	4.021 (3.952)	.0001 (.0001)
Leverage	.527 (.490)	.473 (.454)	.473 (.450)	.467 (.446)	.448 (.437)	.444 (.443)	.450 (.453)	.442 (.444)	.460 (.450)	.508 (.479)	.0428 (.0435)
Prior Year Return	-.055 (-.0193)	.024 (-.057)	.009 (-.057)	.021 (-.036)	.024 (-.023)	.023 (-.020)	.033 (-.016)	.050 (-.025)	.060 (-.035)	.152 (-.050)	.0001 (.0001)
Earnings to Price Ratio	-.564 (-.130)	-.085 (.018)	-.072 (.040)	-.007 (.050)	-.003 (.056)	.016 (.059)	.015 (.060)	.008 (.053)	-.002 (.048)	-.030 (.042)	.0001 (.0001)
Book to Price Ratio	.211 (.494)	0.688 (.603)	.670 (.574)	.795 (.585)	.666 (.599)	.681 (.602)	.676 (.587)	.670 (.538)	.619 (.475)	.469 (.419)	.1734 (.0001)

Size = natural logarithm of the market value of common equity at year end (Price (#24) X Common Shares Outstanding (#25))

Leverage = [Current liabilities (#5) + Long Term Debt (#9)] / Total assets

Prior Year Return = Buy and hold return from over the prior fiscal year (CRSP)

Earnings/Price = Income Before Extraordinary Items #18 / market value of equity (#24*#25)

Book/Price= book value of equity #60 / market value of equity (#24*#25)

Table 4

Descriptive Statistics on Selected Characteristics for a sample portion based on whether firms are Abnormal Sellers, Buyers, or neither.

Sample Consists of 21678 Firm-years between 1985 and 1996.

Panel A: Component of Earnings

	Abnormal Sell (N=1952)		Neither (N=18,141)		Abnormal Buy (N=1585)		Abn. Sell = Neither (p-level)		Abn. Buy = Neither (p-level)		Abn. Sell = Abn. Buy (p-level)	
	<u>Mean</u>	<u>Median</u>	<u>Mean</u>	<u>Median</u>	<u>Mean</u>	<u>Median</u>	<u>T-Test</u>	<u>Median Test</u>	<u>T-Test</u>	<u>Median Test</u>	<u>T-Test</u>	<u>Median Test</u>
Total Accruals	-0.013	-0.022	-0.039	-0.040	-0.045	-0.043	0.001	0.001	0.046	0.018	0.001	0.001
Current Accruals	0.039	0.027	0.013	0.007	0.007	0.004	0.001	0.001	0.015	0.004	0.001	0.001
Cash Flows	0.097	0.135	0.072	0.111	0.064	0.110	0.001	0.001	0.178	0.128	0.001	0.001
Earnings	0.084	0.117	0.033	0.074	0.019	0.068	0.001	0.001	0.019	0.001	0.001	0.001
Jones Disc. Total Accruals	-0.001	0.003	-0.006	-0.002	-0.008	-0.003	0.045	0.011	0.426	0.206	0.046	0.006
DSS Disc. Total Accruals	0.007	0.006	-0.004	-0.001	-0.008	-0.003	0.001	0.001	0.203	0.079	0.001	0.001
Jones Disc. Current Accruals	0.008	0.007	0.001	0.001	-0.001	-0.001	0.003	0.001	0.548	0.384	0.011	0.001
DSS Disc. Current Accruals	0.018	0.013	0.004	0.002	0.001	0.001	0.001	0.001	0.227	0.123	0.001	0.001

Panel B: Financial and Market Characteristics

Size	5.184	5.279	4.732	4.597	4.796	4.640	0.001	0.001	0.243	0.260	0.001	0.001
Leverage	0.429	0.408	0.471	0.453	0.491	0.473	0.001	0.001	0.001	0.001	0.001	0.001
Prior Year Return	0.116	-0.031	0.027	-0.046	0.026	-0.053	0.002	0.036	0.956	0.125	0.002	0.011
Earnings to Price Ratio	-0.011	0.040	-0.076	0.045	-0.058	0.044	0.001	0.006	0.295	0.178	0.001	0.565
Book to Price Ratio	0.425	0.360	0.902	0.568	0.742	0.630	0.085	0.001	0.565	0.001	0.001	0.001

Net Insider Trading Shares (*NITS*) for each firm is measured as $(\text{Shares Purchased}_{i,t} / \text{Shout}_i) - (\text{Shares Sold}_{i,t} / \text{Shout}_i)$ where shares are purchased or sold by executive *i* on day *t*, and normalized by the number of shares outstanding on the transaction date.

Abnormal insider trading is measured relative to the level of insider trading in the same year by firms of equal size (market capitalization decile).

Abnormal Selling = 1 if: (1) *NITS* < 0 and the firm is in size decile 1 or 2, or (2) *NITS* is less than the median of all net selling firms of similar market value, and zero otherwise. *Abnormal Buying* = 1 if: (1) *NITS* > 0 and the firm is in size decile 1 or 2, or (2) *NITS* is greater than the median of all net buying firms of similar market value, and zero otherwise. All other variables are defined as in Table 2.

Table 5
Earnings Persistence of Cash Flow and Accruals Conditional on Insider Trading
Dependent Variable is Operating Income_{t+1}
Sample Consists of 21,678 Firm-years from 1985-1996
(t-statistics in parenthesis)

	(1)	(2)	(3)	(4)	(5)
Intercept	-0.004 (4.40)**	-0.003 (2.88)**	-0.004 (2.88)**	-0.003 (2.47)**	-0.017 (11.5)**
Operating Cash Flow	0.860 (218.5)**	0.860 (214.3)**	0.861 (214.2)**	0.861 (214.2)**	0.848 (204.4)**
Total Accruals	0.752 (101.1)**				
Positive Accruals		0.743 (54.1)**	0.761 (50.4)**	0.761 (50.5)**	
Positive Accruals*Abnormal Selling			-0.170 (4.53)**	-0.171 (4.56)**	
Positive Accruals*Abnormal Buying			0.141 (2.55)**	0.140 (2.53)**	
Negative Accruals		0.757 (72.6)**	0.757 (72.7)**	0.758 (72.8)**	
Positive Current Accruals					0.750 (56.1)**
Positive Current Accruals*Abnormal Selling					-0.141 (4.10)**
Positive Current Accruals*Abnormal Buying					0.099 (2.05)*
Negative Current Accruals					0.758 (57.6)**
Depreciation					-0.465 (24.7)**
Growth (Book to Price Ratio)				-0.001 (2.30)*	-0.001 (2.37)*
Abnormal Insider Selling			0.001 (0.07)	0.001 (0.03)	0.001 (0.26)
Abnormal Insider Buying			0.001 (0.18)	0.001 (0.20)	-0.001 (0.04)

Accruals are calculated as the change in current assets (COMPUSTAT item #4), minus the change in cash (#1), minus changes in current liabilities (#5), plus the change in the current portion of long term debt (#34) minus depreciation and amortization expense (#14) plus change in taxes payable (#71). If items #34 or #71 are missing, then the item is set equal to zero in calculating total accruals since those numbers are likely immaterial for the firm in question.

Operating Cash Flow = (Operating Income after depreciation (#178) - Accruals) / Average total assets (#6).

Earnings = Operating Income after depreciation (#178) / Average total assets (#6).

Unexpected total accruals are estimated from two expectations models:

Abnormal insider trading is measured relative to the level of insider trading in the same year by firms of equal size (market capitalization decile). *Abnormal Selling* = 1 if: (1) insiders are net sellers and the firm is in size decile 1 or 2, or (2) net insider selling exceeds the median for all net selling firms and the sample firm is in size decile 3-10, and zero otherwise.

Abnormal Buying = 1 if: (1) insiders are net buyers and the firm is in size decile 9 or 10, or (2) net insider buying exceeds the median for all net buying firms and the sample firm is in size decile 1-8, and zero otherwise.

Growth = book to price ratio = Book value of equity (#60) / market value of equity (#24*#25)

Table 6
Earnings Persistence of Cash Flow and Unexpected Accruals Conditional on Insider Trading
Dependent Variable is Operating Income_{t+1}
Sample Consists of 21,678 Firm-years from 1985-1996
(t-statistics in parenthesis)

Expectation Model	DSS Current Accruals	Jones Current Accruals	DSS Total Accruals	Jones Total Accruals
Intercept	-0.042 (33.5)**	-0.042 (34.0)**	-0.005 (3.64)**	-0.005 (3.78)**
Operating Cash Flow	0.854 (203.3)**	0.854 (202.8)**	0.862 (210.8)**	0.862 (210.6)**
Expected Accruals	0.761 (46.0)**	0.760 (53.2)**	0.718 (52.3)**	0.723 (59.2)**
Positive Unexpected Accruals	0.761 (52.1)**	0.757 (48.8)**	0.763 (52.6)**	0.764 (50.2)**
Positive Unexpected Accruals*Abnormal Selling	-0.176 (4.47)**	-0.150 (3.45)**	-0.166 (4.21)**	-0.133 (3.11)**
Positive Unexpected Accruals*Abnormal Buying	0.215 (4.71)**	0.217 (4.76)**	0.186 (4.42)**	0.178 (4.27)**
Negative Unexpected Accruals	0.786 (60.8)**	0.787 (60.2)**	0.754 (66.4)**	0.755 (65.4)**
Growth (book to Market Ratio)	-0.001 (1.44)	-0.001 (0.93)	-0.001 (2.20)	-0.001 (2.03)
Abnormal Insider Selling	0.001 (0.35)	-0.001 (0.25)	0.001 (0.24)	-0.001 (0.30)
Abnormal Insider Buying	-0.002 (0.61)	-0.002 (0.50)	-0.002 (0.53)	-0.001 (0.41)

Total accruals are calculated as the change in current assets (COMPUSTAT item #4), minus the change in cash (#1), minus changes in current liabilities (#5), plus the change in the current portion of long term debt (#34) minus depreciation and amortization expense (#14) plus change in taxes payable (#71). If items #34 or #71 are missing, then the item is set equal to zero in calculating total accruals since those numbers are likely immaterial for the firm in question. *Operating Cash Flow* = (Operating Income after depreciation (#178) - Accruals)/Average total assets(#6). *Earnings* = Operating Income after depreciation (#178)/Average total assets(#6). *Unexpected total accruals* are estimated from two expectations models:

(1) Jones' (1991) Model: The model relates total accruals (TACC) to the change in sales ($\Delta Sales_{it}$) and the level of gross property, plant and equipment (PPE #8). $TACC_{it} = \alpha_{1i} + \beta_{1i} \Delta Sales_{it} + \gamma_{1i} PPE_{it} + \varepsilon_{1it}$.

(2) DSS modification: The Jones' (1991) model as modified by (Dechow Sloan and Sweeney 1995, [DSS]) relates total accruals to the change in sales less changes in receivables (#2) ($\Delta Sales_{it} - \Delta Receivables_{it}$) and the level of gross PPE.

$$TACC_{it} = \alpha_{1i} + \beta_{1i} (\Delta Sales_{it} - \Delta Receivables_{it}) + \gamma_{1i} PPE_{it} + \varepsilon_{1it}$$

Unexpected current accruals are estimated from similar expectations models. That is, the dependent variable is current accruals (CACC) which equals TACC + Depreciation, and the models have just one explanatory variable since the PPE variable is dropped.

Abnormal insider trading is measured relative to the level of insider trading in the same year by firms of equal size (market capitalization decile). *Abnormal Selling* = 1 if: (1) insiders are net sellers and the firm is in size decile 1 or 2, or (2) net insider selling exceeds the median for all net selling firms and the sample firm is in size decile 3-10, and zero otherwise.

Abnormal Buying = 1 if: (1) insiders are net buyers and the firm is in size decile 9 or 10, or (2) net insider buying exceeds the median for all net buying firms and the sample firm is in size decile 1-8, and zero otherwise.

Growth = book to price ratio = Book value of equity (#60) / market value of equity (#24*#25)

Table 7
Tests on the Rational Pricing of Earnings Components Conditional on Insider Trading
Sample Consists of 21,678 Firm-years from 1985-1996

Panel A - Cash Flow and Accruals

$$\text{Earnings}_{t+1} = a1 \cdot \text{CFO} + a2 \cdot \text{ACCRUAL} + \varepsilon_{t+1}$$

$$\text{Size Adjusted Return}_{t+1} = \beta(\text{Earnings}_{t+1} - \alpha0 - \alpha1 \cdot \text{CFO} - \alpha2 \cdot \text{ACCRUAL}) + e_{t+1}$$

Earnings Expectations Regression			Pricing Regression			^(a) Test of Equality $a_i = \alpha_i$	
Parameter	Estimate	Standard Error	Parameter	Estimate	Standard Error	Likelihood Ratio	p-value
a1	0.860	0.0039	$\alpha1$	0.749	0.0192	33.17	0.001
a2	0.752	0.0074	$\alpha2$	0.874	0.0359	11.39	0.001

Panel B - Cash Flow, Positive, and Negative Accruals

$$\text{Earnings}_{t+1} = a1 \cdot \text{CFO} + a2 \cdot \text{POSACC} + a3 \cdot \text{NEGACC} + \varepsilon_{t+1}$$

$$\text{Size Adjusted Return}_{t+1} = \beta(\text{Earnings}_{t+1} - \alpha0 - \alpha1 \cdot \text{CFO} - \alpha2 \cdot \text{POSACC} - \alpha3 \cdot \text{NEGACC}) + e_{t+1}$$

Earnings Expectations Regression			Pricing Regression			^(a) Test of Equality $a_i = \alpha_i$	
Parameter	Estimate	Standard Error	Parameter	Estimate	Standard Error	Likelihood Ratio	p-value
a1	0.860	0.0040	$\alpha1$	0.758	0.0196	26.57	0.001
a2	0.743	0.0137	$\alpha2$	1.011	0.0666	15.86	0.001
a3	0.757	0.0104	$\alpha3$	0.788	0.0501	0.36	0.549

Panel C - Cash Flow, Accruals, and Insider Trading ^(b)

$$\text{Earnings}_{t+1} = a1 \cdot \text{CFO} + a2 \cdot \text{POSACC} + a3 \cdot \text{POSACC} \cdot \text{SELL} + a4 \cdot \text{POSACC} \cdot \text{BUY} + a5 \cdot \text{NEGACC} + a6 \cdot \text{SELL} + a7 \cdot \text{BUY} + a8 \cdot \text{BP} + \varepsilon_{t+1}$$

$$\text{Size Adjusted Return}_{t+1} = \beta(\text{Earnings}_{t+1} - \alpha0 - \alpha1 \cdot \text{CFO} - \alpha2 \cdot \text{POSACC} - \alpha3 \cdot \text{POSACC} \cdot \text{SELL} - \alpha4 \cdot \text{POSACC} \cdot \text{BUY} - \alpha5 \cdot \text{NEGACC} - \alpha6 \cdot \text{SELL} - \alpha7 \cdot \text{BUY} - \alpha8 \cdot \text{BP}) + e_{t+1}$$

Earnings Expectations Regression			Pricing Regression			Test of Equality $a_i = \alpha_i$	
Parameter	Estimate	Standard Error	Parameter	Estimate	Standard Error	Likelihood Ratio	p-value
a1	0.860	0.0040	$\alpha1$	0.754	0.0196	29.13	0.001
a2	0.761	0.0151	$\alpha2$	0.961	0.0729	7.35	0.007
a3	-0.171	0.0375	$\alpha3$	0.235	0.1807	4.87	0.014
a4	0.140	0.0553	$\alpha4$	0.056	0.2658	0.09	0.759
a5	0.759	0.0104	$\alpha5$	0.789	0.0502	0.36	0.548
a6	0.001	0.0032	$\alpha6$	0.025	0.0155	2.58	0.108
a7	0.001	0.0034	$\alpha7$	-0.001	0.0163	0.01	0.922
a8	-0.001	0.0003	$\alpha8$	-0.003	0.0012	4.56	0.033

^(a) Hypothesis tests: (insider selling) $a2 + a3 = \alpha2 + \alpha3$ Likelihood Ratio = 12.44 p-value= 0.001
(insider buying) $a2 + a4 = \alpha2 + \alpha4$ Likelihood Ratio = 0.20 p-value= 0.657

Panel D - Cash Flow, Current Accruals, and Insider Trading ^(b)

$$\text{Earnings}_{t+1} = a1 \cdot \text{CFO} + a2 \cdot \text{POSACC} + a3 \cdot \text{POSACC} \cdot \text{SELL} + a4 \cdot \text{POSACC} \cdot \text{BUY} + a5 \cdot \text{NEGACC} + a6 \cdot \text{SELL} + a7 \cdot \text{BUY} + a8 \cdot \text{DEP} + a9 \cdot \text{BP} + \varepsilon_{t+1}$$

$$\text{Size Adjusted Return}_{t+1} = \beta(\text{Earnings}_{t+1} - \alpha0 - \alpha1 \cdot \text{CFO} - \alpha2 \cdot \text{POSACC} - \alpha3 \cdot \text{POSACC} \cdot \text{SELL} - \alpha4 \cdot \text{POSACC} \cdot \text{BUY} - \alpha5 \cdot \text{NEGACC} - \alpha6 \cdot \text{SELL} - \alpha7 \cdot \text{BUY} - \alpha8 \cdot \text{DEP} - \alpha9 \cdot \text{BP}) + e_{t+1}$$

Earnings Expectations Regression			Pricing Regression			Test of Equality $a_i = \alpha_i$	
Parameter	Estimate	Standard Error	Parameter	Estimate	Standard Error	Likelihood Ratio	p-value
a1	0.848	0.0042	$\alpha1$	0.749	0.0202	23.79	0.001
a2	0.750	0.0133	$\alpha2$	0.920	0.0641	6.82	0.009
a3	-0.141	0.0344	$\alpha3$	0.274	0.1655	6.08	0.007
a4	0.099	0.0482	$\alpha4$	0.060	0.2315	0.03	0.872
a5	0.758	0.0132	$\alpha5$	0.800	0.0634	0.43	0.511
a6	0.001	0.0036	$\alpha6$	0.015	0.0171	0.69	0.405
a7	-0.001	0.0037	$\alpha7$	-0.001	0.0176	0.04	0.840
a8	-0.465	0.0188	$\alpha8$	-0.579	0.0905	1.53	0.217
a9	-0.001	0.0003	$\alpha9$	-0.003	0.0012	4.98	0.026

^(a) Hypothesis tests: (insider selling) $a2 + a3 = \alpha2 + \alpha3$ Likelihood Ratio = 13.61 p-value= 0.001
(insider buying) $a2 + a4 = \alpha2 + \alpha4$ Likelihood Ratio = 0.33 p-value= 0.566

Table 7 - Continued
Tests on the Rational Pricing of Unexpected Accruals Conditional on Insider Trading
Sample Consists of 21,678 Firm-years from 1985-1996

Panel E - Cash Flow, Unexpected Current Accruals, and Insider Trading ^{(b) (c)}							
Earnings _{t+1} = a1*CFO + a2*POSUXC + a3*POSUXC*SELL + a4*POSUXC*BUY + a5*NEGUXC + a6*EACC + a7*SELL + a8*BUY + a9*BP + ε _{t+1}							
Size Adjusted Return _{t+1} = β(Earnings _{t+1} - α0 - α1*CFO - α2*POSUXC - α3*POSUXC*SELL - α4*POSUXC*BUY - α5*NEGUXC - α6*EACC - α7*SELL - α8*BUY - α9*BP) + e _{t+1}							
Earnings Expectations Regression			Pricing Regression			Test of Equality ai = αi	
Parameter	Estimate	Standard Error	Parameter	Estimate	Standard Error	Likelihood Ratio	p-value
a1	0.854	0.0042	α1	0.746	0.0210	26.33	0.001
a2	0.761	0.0146	α2	0.958	0.0723	7.26	0.007
a3	-0.176	0.0394	α3	0.245	0.1951	4.51	0.017
a4	0.215	0.0456	α4	-0.079	0.2254	1.64	0.201
a5	0.786	0.0129	α5	0.849	0.0639	0.93	0.336
a6	0.760	0.0165	α6	0.884	0.0816	2.21	0.137
a7	0.001	0.0035	α7	0.023	0.0173	1.54	0.214
a8	-0.002	0.0036	α8	-0.001	0.0180	0.01	0.922
a9	-0.001	0.0003	α9	-0.003	0.0013	5.07	0.024
^(a) Hypothesis tests: (insider selling) a2 + a3 = α2 + α3 Likelihood Ratio = 10.86 p-value= 0.001							
(insider buying) a2 + a4 = α2 + α4 Likelihood Ratio = 0.18 p-value= 0.668							

POSACC (NEGACC) = Total accruals when total accruals are positive (negative), and zero otherwise

POSCA (NEGCA) = Current accruals when current accruals are positive (negative), and zero otherwise.

POSUXC (NEGUXC) = Unexpected current accruals when unexpected current accruals are positive (negative), and zero otherwise.

EACC = Expected Current Accruals. DEP = Depreciation and Amortization. CFO = Operating Cash Flow.

BP = Book value of equity / Market value of equity

(a) - From Mishkin (1983) the test statistic $2*N*\log(SSR^c/SSR^u)$ is asymptotically $\chi^2(q)$ distributed where:
q = the number of rational pricing constraints imposed, N = the number of observations in the sample
SSR^c = the sum of squared residuals from the constrained regressions in the second stage,
SSR^u = the sum of squared residuals from the unconstrained regressions in the first stage.

(b) - Abnormal insider trading is measured relative to the level of insider trading in the same year by firms of equal size (market capitalization decile). Sell = 1 if: (1) insiders are net sellers and the firm is in size decile 1 or 2, or (2) net insider selling exceeds the median for all net selling firms and the sample firm is in size decile 3-10, and zero otherwise.

Buy = 1 if: (1) insiders are net buyers and the firm is in size decile 9 or 10, or (2) net insider buying exceeds the median for all net buying firms and the sample firm is in size decile 1-8, and zero otherwise.

(c) - Unexpected current accruals are estimated from the Jones' (1991) model as modified by (Dechow Sloan and Sweeney (1995).

Table 8
Tests on the Rational Pricing of Unexpected Accruals conditional on Insider Trading
Insider Consensus - Unanimous Buying or Selling
Sample Consists of 21,678 Firm-years from 1985-1996

Panel A - Cash Flow, Current Accruals, and Insider Trading^(b)

$$\text{Earnings}_{t+1} = a1 \cdot \text{CFO} + a2 \cdot \text{POSCA} + a3 \cdot \text{POSCA} \cdot \text{SELL} + a4 \cdot \text{POSCA} \cdot \text{BUY} + a5 \cdot \text{NEGCA} + a6 \cdot \text{SELL} \\ + a7 \cdot \text{BUY} + a8 \cdot \text{DEP} + \varepsilon_{t+1}$$

$$\text{Size Adjusted Return}_{t+1} = \beta(\text{Earnings}_{t+1} - \alpha0 - \alpha1 \cdot \text{CFO} - \alpha2 \cdot \text{POSCA} - \alpha3 \cdot \text{POSCA} \cdot \text{SELL} - \alpha4 \cdot \text{POSCA} \cdot \text{BUY} \\ - \alpha5 \cdot \text{NEGCA} - \alpha6 \cdot \text{SELL} - \alpha7 \cdot \text{BUY} - \alpha8 \cdot \text{DEP}) + e_{t+1}$$

Earnings Expectations Regression			Pricing Regression			^(a) Test of Equality $a_i = \alpha_i$	
Parameter	Estimate	Standard Error	Parameter	Estimate	Standard Error	Likelihood Ratio	p-value
a1	0.847	0.0042	$\alpha1$	0.746	0.0202	24.66	0.001
a2	0.752	0.0142	$\alpha2$	0.907	0.0681	5.01	0.025
a3	-0.090	0.0294	$\alpha3$	0.214	0.1412	4.43	0.017
a4	0.006	0.0368	$\alpha4$	0.126	0.1763	0.45	0.503
a5	0.757	0.0132	$\alpha5$	0.788	0.0634	0.24	0.627
a6	0.002	0.0030	$\alpha6$	0.018	0.0128	1.51	0.219
a7	-0.002	0.0027	$\alpha7$	-0.024	0.0144	2.31	0.128
a8	-0.463	0.0188	$\alpha8$	-0.568	0.0904	1.31	0.253

^(a) Hypothesis tests: $a2 + a3 = \alpha2 + \alpha3$ Likelihood Ratio = 12.42 p-value= 0.001

$a2 + a4 = \alpha2 + \alpha4$ Likelihood Ratio = 2.63 p-value= 0.106

Panel B - Cash Flow, Unexpected Current Accruals, and Insider Trading^{(b)(c)}

$$\text{Earnings}_{t+1} = a1 \cdot \text{CFO} + a2 \cdot \text{POSUXC} + a3 \cdot \text{POSUXC} \cdot \text{SELL} + a4 \cdot \text{POSUXC} \cdot \text{BUY} + a5 \cdot \text{NEGUXC} + a6 \cdot \text{EACC} \\ + a7 \cdot \text{SELL} + a8 \cdot \text{BUY} + \varepsilon_{t+1}$$

$$\text{Size Adjusted Return}_{t+1} = \beta(\text{Earnings}_{t+1} - \alpha0 - \alpha1 \cdot \text{CFO} - \alpha2 \cdot \text{POSUXC} - \alpha3 \cdot \text{POSUXC} \cdot \text{SELL} - \alpha4 \cdot \text{POSUXC} \cdot \text{BUY} \\ - \alpha5 \cdot \text{NEGUXC} - \alpha6 \cdot \text{EACC} - \alpha7 \cdot \text{SELL} - \alpha8 \cdot \text{BUY}) + e_{t+1}$$

Earnings Expectations Regression			Pricing Regression			^(a) Test of Equality $a_i = \alpha_i$	
Parameter	Estimate	Standard Error	Parameter	Estimate	Standard Error	Likelihood Ratio	p-value
a1	0.854	0.0042	$\alpha1$	0.742	0.0211	27.80	0.001
a2	0.758	0.0154	$\alpha2$	0.922	0.0759	4.49	0.034
a3	-0.083	0.0344	$\alpha3$	0.313	0.1697	5.26	0.011
a4	0.088	0.0368	$\alpha4$	0.085	0.1813	0.01	0.962
a5	0.785	0.0130	$\alpha5$	0.836	0.0638	0.62	0.432
a6	0.754	0.0165	$\alpha6$	0.882	0.0814	2.41	0.120
a7	0.001	0.0027	$\alpha7$	0.020	0.0131	2.18	0.140
a8	-0.005	0.0030	$\alpha8$	-0.024	0.0149	1.68	0.195

^(a) Hypothesis tests: $a2 + a3 = \alpha2 + \alpha3$ Likelihood Ratio = 12.45 p-value= 0.001

$a2 + a4 = \alpha2 + \alpha4$ Likelihood Ratio = 0.88 p-value= 0.349

POSCA (NEGCA) = Current accruals when current accruals are positive (negative), and zero otherwise.

POSUXC (NEGUXC) = Unexpected current accruals when unexpected current accruals are positive (negative), and zero otherwise.

EACC = Expected Current Accruals. *DEP* = Depreciation and Amortization. *CFO* = Operating Cash Flow.

(a) - From Mishkin (1983) the test statistic $2 \cdot N \cdot \log(\text{SSR}^c / \text{SSR}^u)$ is asymptotically $\chi^2(q)$ distributed where:

q = the number of rational pricing constraints imposed, N = the number of observations in the sample

SSR^c = the sum of squared residuals from the constrained regressions in the second stage,

SSR^u = the sum of squared residuals from the unconstrained regressions in the first stage.

(b) - Abnormal insider trading is measured relative to the level of insider trading in the same year by firms of equal size (market capitalization decile). *Sell* = 1 if: (1) insiders are net sellers and the firm is in size decile 1 or 2, or (2) net insider selling exceeds the median for all net selling firms and the sample firm is in size decile 3-10, and zero otherwise. *Buy* = 1 if: (1) insiders are net buyers and the sample firm is in size decile 9 or 10, or (2) net insider buying exceeds the median for all net buying firms and the sample firm is in size decile 1-8.

(c) - Unexpected current accruals are estimated from the Jones' (1991) model as modified by (Dechow Sloan and Sweeney (1995)).

Table 9
Insider Trading and Subsequent Size Adjusted Portfolio Stock Returns
in the Year Following Accrual Portfolio Formations

(p-values in parentheses)
Sample Consists of 21678 Firm-years from 1985-1996

Panel A: Average Returns for Portfolio Assignment Based On Total Accruals				
Portfolio ^(a)	Accruals Only	Accruals and Insider Trading		
		Abnormal Selling ^(b)	Abnormal Buying	No Insider Trading
Lowest	0.055 (0.011)	0.025 (0.687)	0.045 (0.460)	0.059 (0.016)
Highest	-0.055 (0.001)	-0.147 (0.001)	-0.025 (0.594)	-0.039 (0.023)
Hedge ^(d)	(1) High and Low Accruals (Sloan)		0.110	
Returns	(2) High Accruals and Abnormal Selling		0.147	
	(3) High Accruals, Abnormal Selling and High Cash Flow		0.221	
Panel B: Average Returns for Portfolio Assignment Based On Current Accruals				
Portfolio ^(a)	Accruals Only	Accruals and Insider Trading		
		Abnormal Selling ^(b)	Abnormal Buying	No Insider Trading
Lowest	0.068 (0.002)	0.023 (0.702)	0.040 (0.488)	0.075 (0.003)
Highest	-0.056 (0.001)	-0.129 (0.002)	-0.024 (0.636)	-0.044 (0.010)
Hedge ^(d)	(1) High and Low Accruals (Sloan)		0.124	
Returns	(2) High Accruals and Abnormal Selling		0.129	
	(3) High Accruals, Abnormal Selling and High Cash Flow		0.204	
Panel C: Average Returns for Portfolio Assignment Based On Unexpected Current Accruals ^(c)				
Portfolio ^(a)	Accruals Only	Accruals and Insider Trading		
		Abnormal Selling ^(b)	Abnormal Buying	No Insider Trading
Lowest	0.063 (0.004)	0.034 (0.560)	-0.045 (0.354)	0.077 (0.003)
Highest	-0.059 (0.001)	-0.140 (0.006)	-0.013 (0.784)	-0.049 (0.005)
Hedge ^(d)	(1) High and Low Accruals (Sloan)		0.122	
Returns	(2) High Accruals and Abnormal Selling		0.140	
	(3) High Accruals, Abnormal Selling and High Cash Flow		0.215	

Returns are size adjusted, computed as the difference between the 12-month buy and hold return on a portfolio starting four months after the fiscal year end in which accruals are measured and the 12-month buy and hold return on a value weighted portfolio of similar sized (market value) firms.

(a) - Firms are ranked according to the magnitude of their accruals and assigned in equal numbers to ten portfolios each year. The highest accrual decile corresponds to the most positive (income increasing) accruals.

(b) - Abnormal insider trading is measured relative to the level of insider trading in the same year by firms of equal size (market capitalization decile). *Abnormal Selling* = 1 if: (1) insiders are net sellers and the firm is in size decile 1 or 2, or (2) net insider selling exceeds the median for all net selling firms and the sample firm is in size decile 3-10, and zero otherwise.

Abnormal Buying = 1 if: (1) insiders are net buyers and the firm is in size decile 9 or 10, or (2) net insider buying exceeds the median for all net buying firms and the sample firm is in size decile 1-8, and zero otherwise.

(c) - Expected current accruals are estimated from the Jones' (1991) model as modified by Dechow, et.al. (1995).

(d) - *The High and Low Accrual* portfolio consists of a long position in the lowest accrual decile and an offsetting short position in the highest decile. *The High Accruals and Abnormal Selling* portfolio consists of a long position in firms from the highest accrual decile which exhibit abnormal selling, and an offsetting position in firms of similar size.

The High Accruals, Abnormal Selling and High Cash Flow Hedge consist of a long position in firms from the highest accrual decile which exhibit abnormal selling, and an offsetting position in firms from the highest cash flow decile. The average size adjusted return to the high cash flow decile was 7.48 percent (t=5.42)

Table 10
Hedge Portfolio Performance After Controlling for Various Factors
Sample Consists of 21678 Firm-years from 1985-1996
Dependent Variable is Size Adjusted Return $t+1$ (p-level in parentheses)

Model:	<i>Accruals Only</i> ^(a)			(4)	<i>Accruals, Cash Flow, and Insider Trading</i> ^(b)		
	(1) Total Accruals	(2) Current Accruals	(3) Unexp. Current Accruals		(5) Insider Trading & Current Accruals	(6) Insider Trading & Current Accruals	(7) IT & Unexp. Current Accruals
Long Portfolio:	Accrual Decile 1	Accrual Decile 1	Accrual Decile 1	Abnormal Buying	Firms of Equal Market Value	Cash Flow Decile 10	Cash Flow Decile 10
Short Portfolio:	Accrual Decile 10	Accrual Decile 10	Accrual Decile 10	Abnormal Selling	Accrual Decile 10 and Abnormal Selling	Accrual Decile 10 and Abnormal Selling	Accrual Decile 10 and Abnormal Selling
Intercept	-0.054 (0.003)	-0.054 (0.005)	-0.056 (0.004)	-0.006 (0.713)		-0.138 (0.001)	-0.152 (0.001)
Hedge	0.109 (0.001)	0.119 (0.001)	0.118 (0.001)	0.028 (0.282)	0.129 (0.005)	0.215 (0.001)	0.229 (0.001)
BP	0.002 (0.468)	0.002 (0.394)	0.009 (0.132)	0.036 (0.074)	-0.129 (0.478)	-0.005 (0.245)	-0.005 (0.253)
RET(t-1)	0.026 (0.068)	0.012 (0.403)	0.008 (0.583)	0.010 (0.505)	-0.008 (0.828)	0.008 (0.632)	0.012 (0.510)
EPRATIO	0.018 (0.038)	0.018 (0.094)	0.007 (0.562)	-0.170 (0.001)	-0.365 (0.467)	0.036 (0.154)	0.036 (0.162)
LMVE	-0.022 (0.004)	-0.020 (0.010)	-0.020 (0.014)	-0.009 (0.178)	-0.022 (0.471)	-0.027 (0.001)	-0.029 (0.001)
SURP	0.003 (0.661)	0.007 (0.410)	0.013 (0.196)	0.202 (0.001)	-0.219 (0.671)	-0.002 (0.794)	-0.002 (0.796)
Number of Observations	4251	4257	4117	3537	337	2480	2428

HEDGE=1 for firms where the trading strategy dictates a long position, 0 otherwise. *BP* = ratio of book value of equity to market value of equity.

RET(t-1) = buy and hold return from over the prior fiscal year (CRSP). *EPRATIO* = earnings before extraordinary items / market value of equity.

LMVE = natural logarithm of the market value of common equity at year end *SURP* = the difference between income before extraordinary items in the fourth quarter in the most recent year and the same quarter in the previous year, deflated by market value of equity at the end of the most recent fiscal year

Variables *BP*, *PRIORET*, *EPRATIO*, and *LMVE* are measured in the fiscal year preceding the year in which the hedge portfolio is formed. All variables (excluding *HEDGE*) are mean adjusted.

(a) - Firms are ranked according to the magnitude of their accruals and assigned in equal numbers to ten portfolios each year. The highest accrual decile corresponds to the most positive (income increasing) accruals. Expected Accruals are forecasted one year ahead using the Dechow, et.al., 1995 modification of the Jones model.

(b) - Insider trading for each firm is measured as $(\text{Shares Purchased}_{i,t} / \text{Shout}_i) - (\text{Shares Sold}_{i,t} / \text{Shout}_i)$ where shares purchased and sold by executive *i* are open market purchases and sales. *Shout* is the number of shares outstanding on the date of the insider transaction (*t*).

Abnormal insider trading is measured relative to the level of insider trading in the same year by firms of equal size (market capitalization decile).

Abnormal Selling = 1 if: (1) insiders are net sellers and the firm is in size decile 1 or 2, or (2) net insider selling exceeds the median for all net selling firms and the sample firm is in size decile 3-10, and zero otherwise.

Abnormal Buying = 1 if: (1) insiders are net buyers and the firm is in size decile 9 or 10, or (2) net insider buying exceeds the median for all net buying firms and the sample firm is in size decile 1-8, and zero otherwise.